I need to know all possible web app tools that AdSense approve and are easy to make for me to get AdSense monetization is up. So, I need to know all possible tools and as an expert in every role needed to make all the tools in each category of any developer, UI UX designer, any kind of role you are in. I need you to focus on what AdSense requires to make those web tools. List the tools first, then as you roll in, give me the full plan to make all each of those tools. Make the prompt better if needed then Give me the answer to everything with every conceivable info that guarantees compliance with all adsense requires and then the implementation as all conceivable roles coming together to achieve it

Complete blueprint: Building an AdSense-ready web tools platform

Your goal should be to build a **web-based tools platform** containing useful, free, fast, original online utilities that attract organic Google traffic and can eventually earn revenue through Google AdSense.

The strongest strategy is not to publish hundreds of tiny calculators and hope Google approves the website. It is to build a genuinely useful platform with carefully selected tools, original supporting content, excellent user experience, transparent ownership, and a technically sound advertising implementation.

I will approach this as a combined team of:

- Product strategist and business analyst
- Google AdSense and search policy specialist
- SEO and content strategist
- Full-stack software architect
- Frontend and backend developer
- UI/UX designer
- Accessibility specialist
- Database and infrastructure engineer
- Privacy and security specialist
- Quality assurance engineer
- Technical SEO specialist
- Conversion and analytics specialist

First: One critical truth about AdSense approval

Important

Nobody can guarantee AdSense approval.

Google evaluates websites and publisher accounts according to its current policies and eligibility requirements. Having a certain number of tools, articles, visitors, pages, or words does not automatically qualify a website.

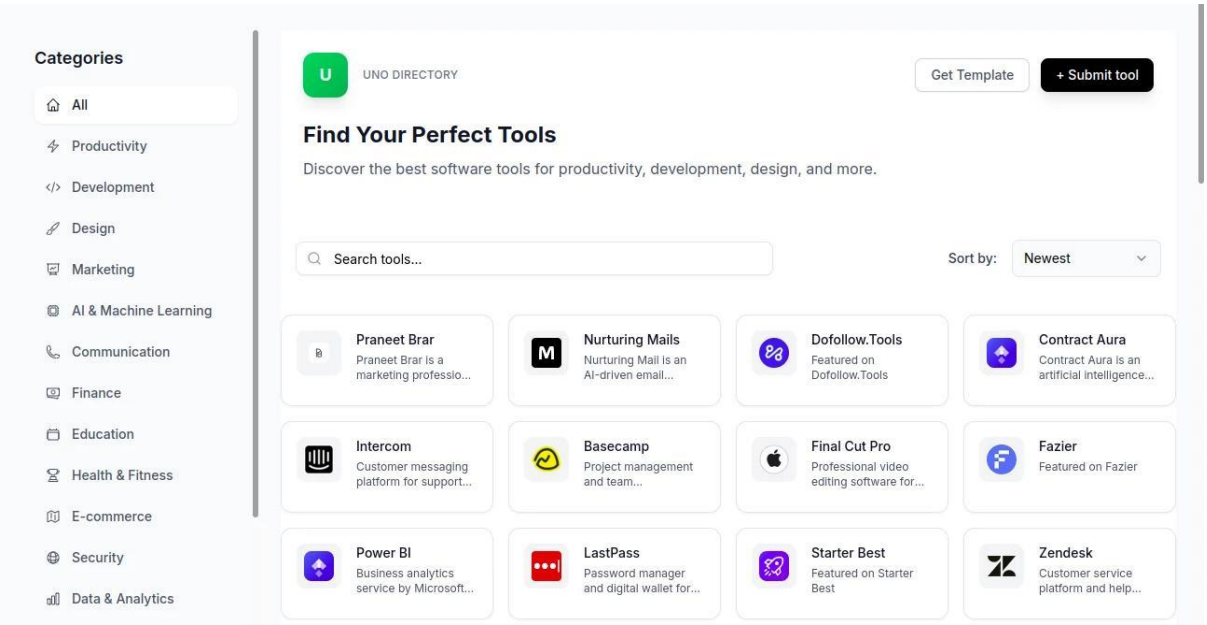
Our objective is to maximize the likelihood of approval by satisfying applicable requirements, avoiding policy violations, and creating a website that provides substantial value to visitors.

We should design the platform to be compliant from the beginning rather than trying to repair it after rejection.

Part 1. The complete web-tool catalogue

I recommend building one website with a clear identity and multiple categories rather than launching many unrelated websites.

Think of it as an online utility centre.



Your Online Tools Hub

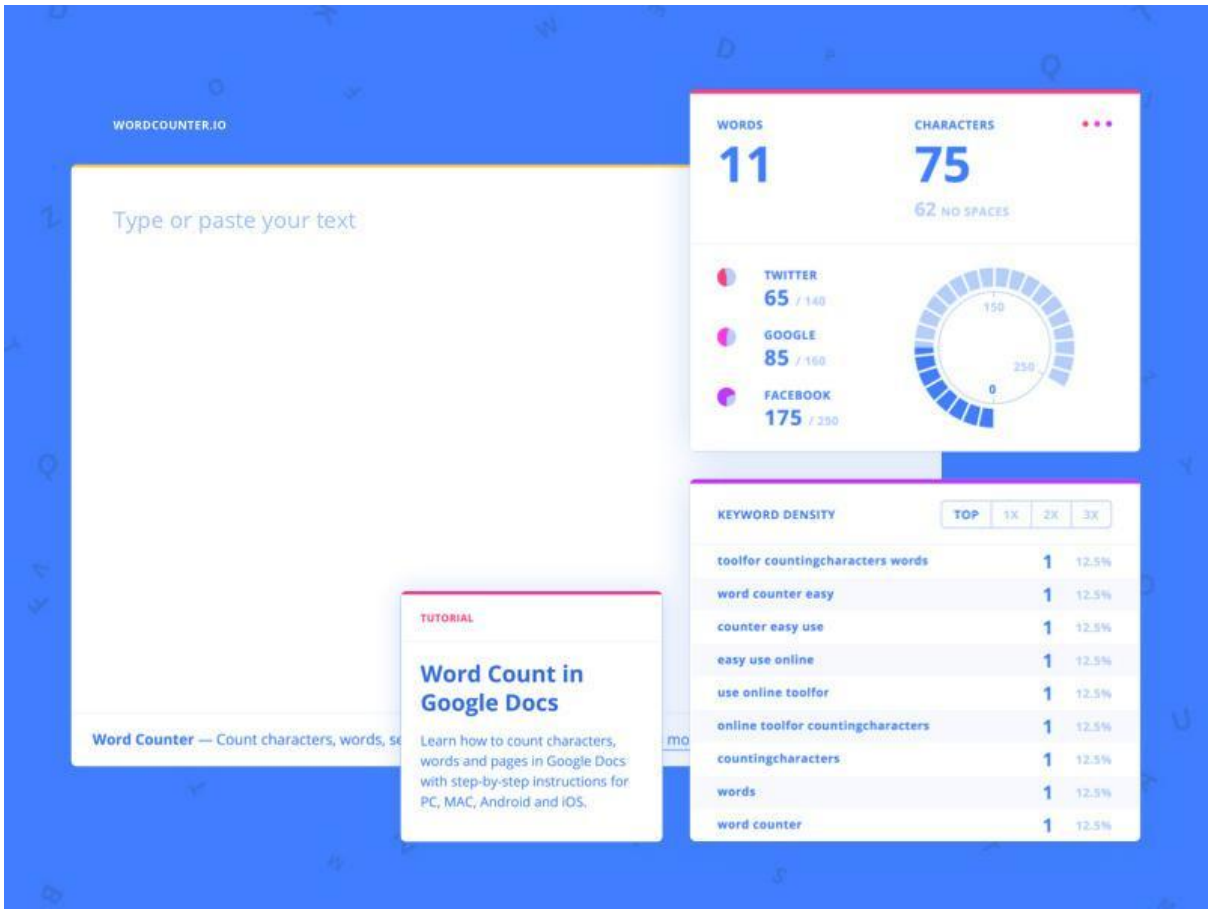
One domain · Multiple categories · Individual landing pages · Free tools · Organic search traffic

The catalogue below is a broad selection of practical tools that can be built without expensive infrastructure. The classifications are my product-development recommendations, not Google's official AdSense approval categories.

A. Text and writing tools

Excellent starting category

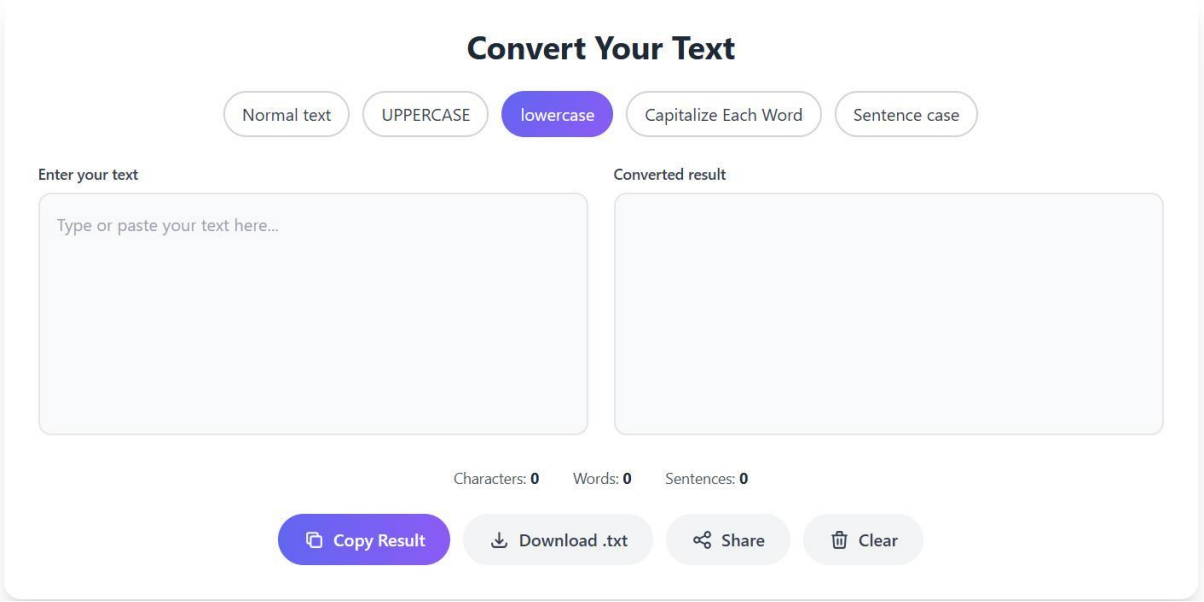
These are relatively easy to build, can provide immediate value, and usually have inexpensive operating costs.



1. Word Counter

Count words, characters, sentences, paragraphs, reading time and speaking time.

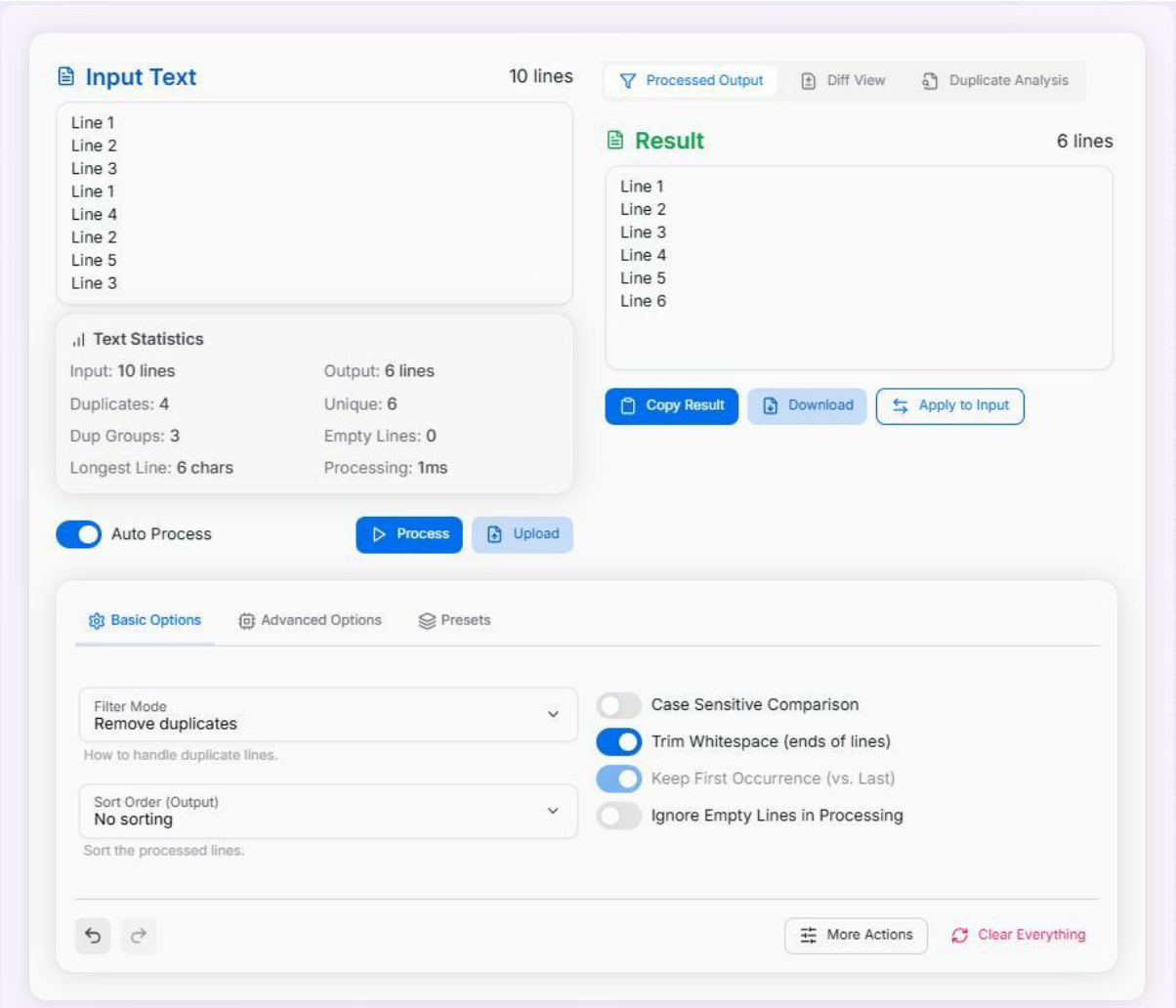
Difficulty: Easy · Cost: Very low



2. Case Converter

Uppercase, lowercase, title case, sentence case and alternating case.

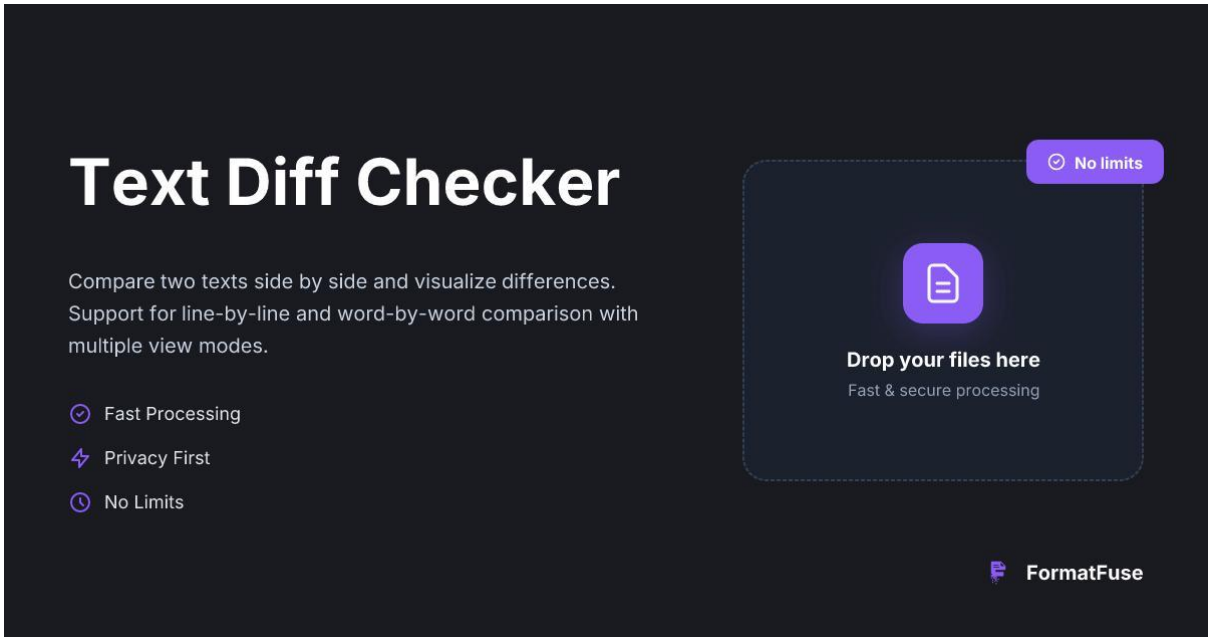
Difficulty: Easy · Cost: Very low



3. Duplicate Line Remover

Remove duplicate lines and retain original ordering.

Difficulty: Easy · Cost: Very low



4. Text Difference Checker

Compare two texts and highlight additions, removals and changes.

Difficulty: Easy · Cost: Very low

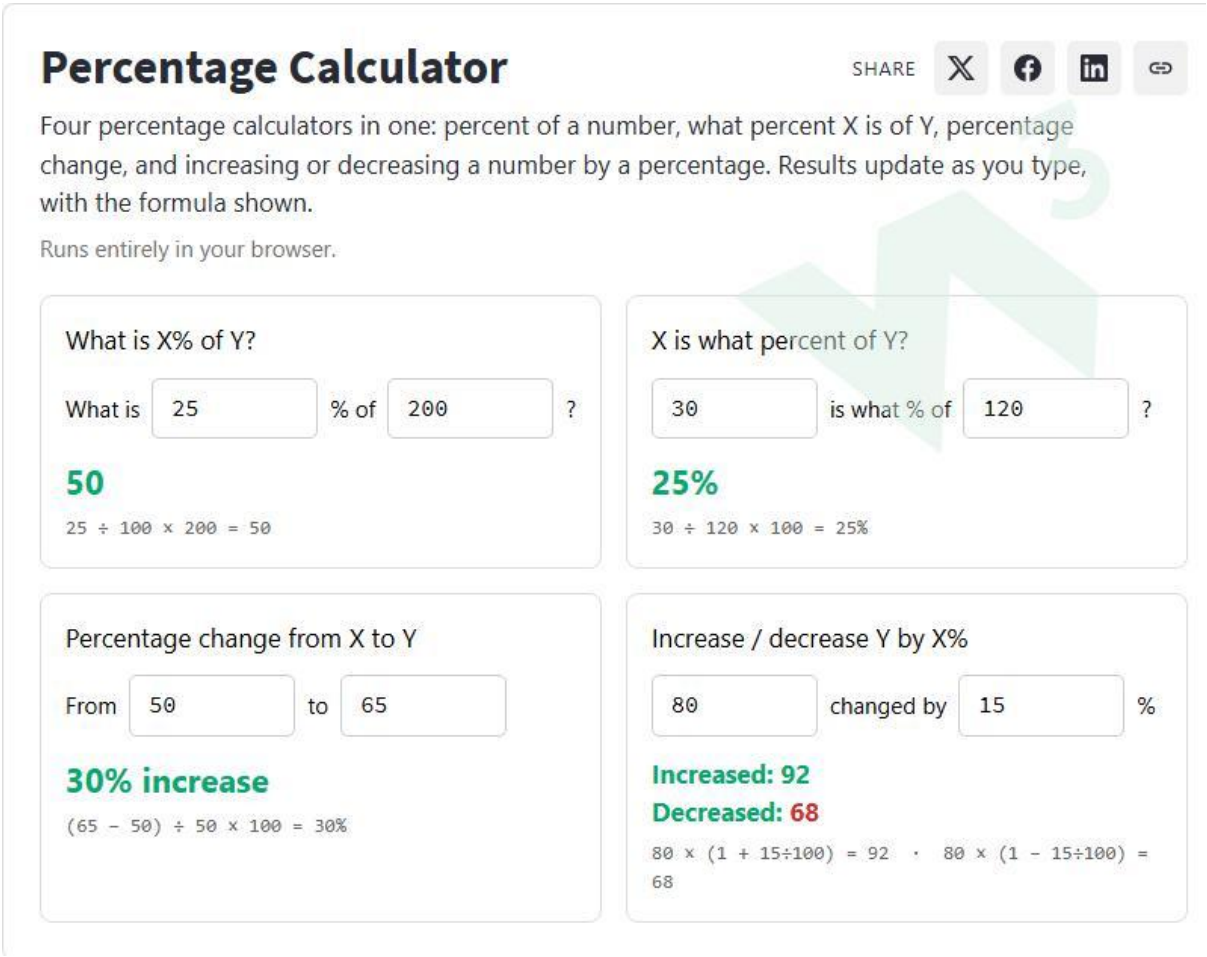
More text tools:

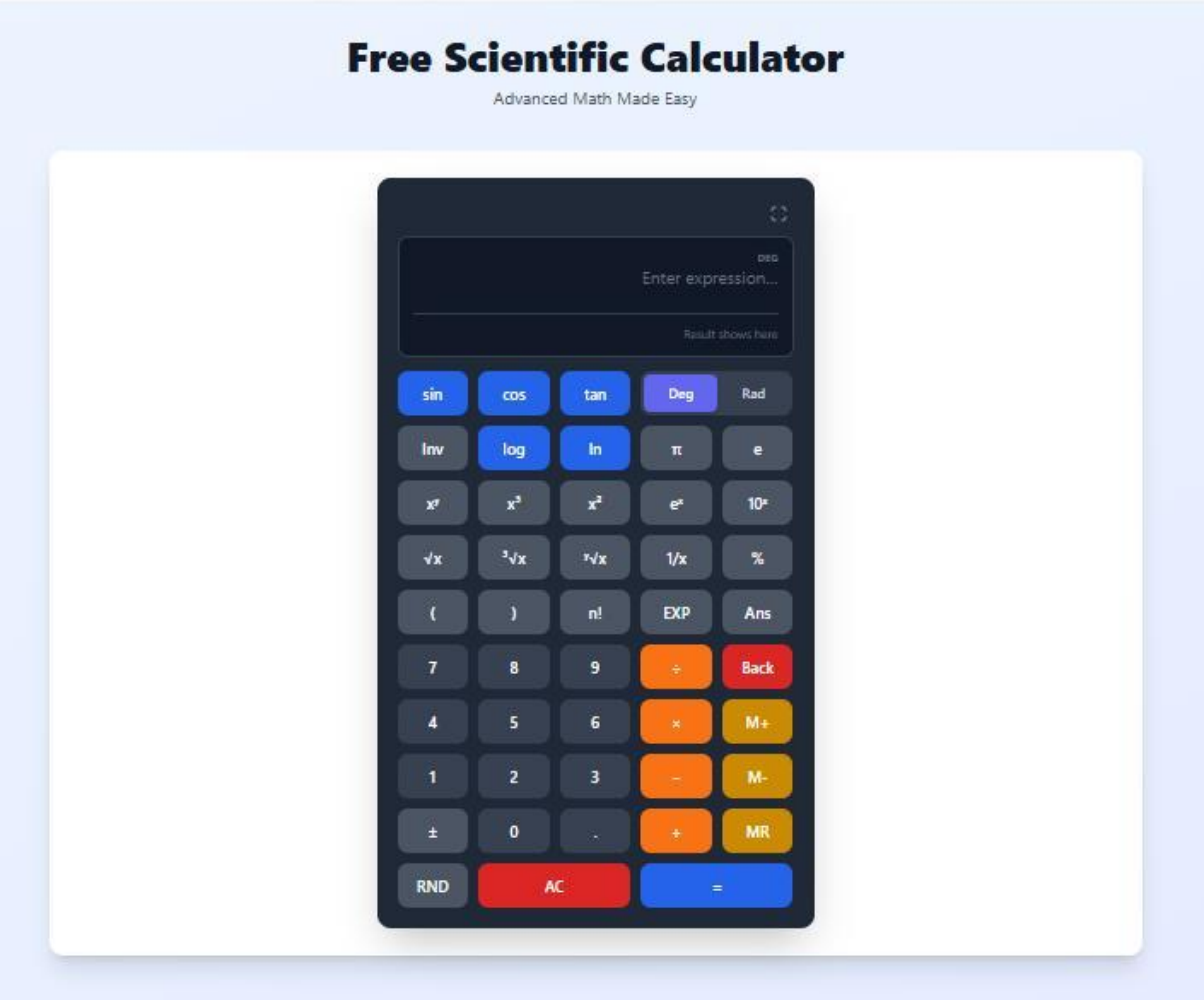
5. Character Counter
6. Sentence Counter
7. Paragraph Counter
8. Reading Time Calculator
9. Speaking Time Calculator
10. Text Reverser
11. Alphabetical Text Sorter
12. Remove Extra Spaces
13. Line Break Remover
14. Text to Slug Converter
15. Slug to Text Converter
16. Find and Replace Tool
17. Text Frequency Analyzer
18. Lorem Ipsum Generator

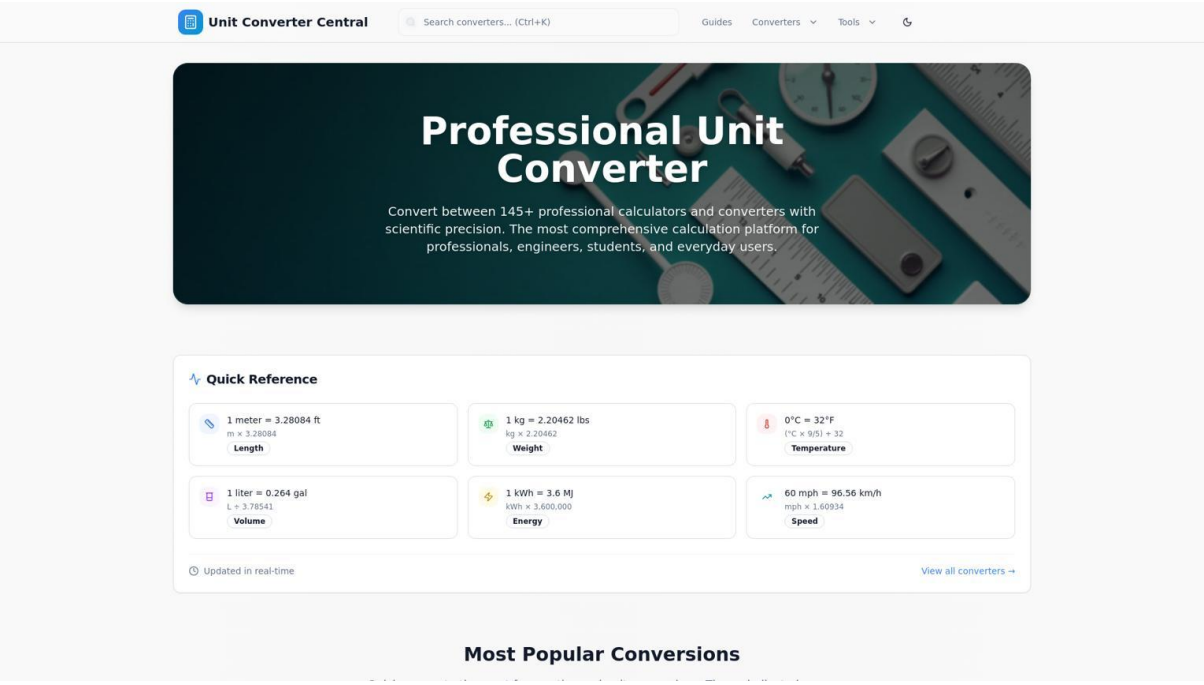
19. Random Text Generator
20. Markdown Previewer
21. Markdown to HTML Converter
22. HTML to Markdown Converter
23. HTML Entity Encoder/Decoder
24. URL Encoder/Decoder
25. Unicode Character Inspector

B. Calculators and mathematics

Excellent starting category







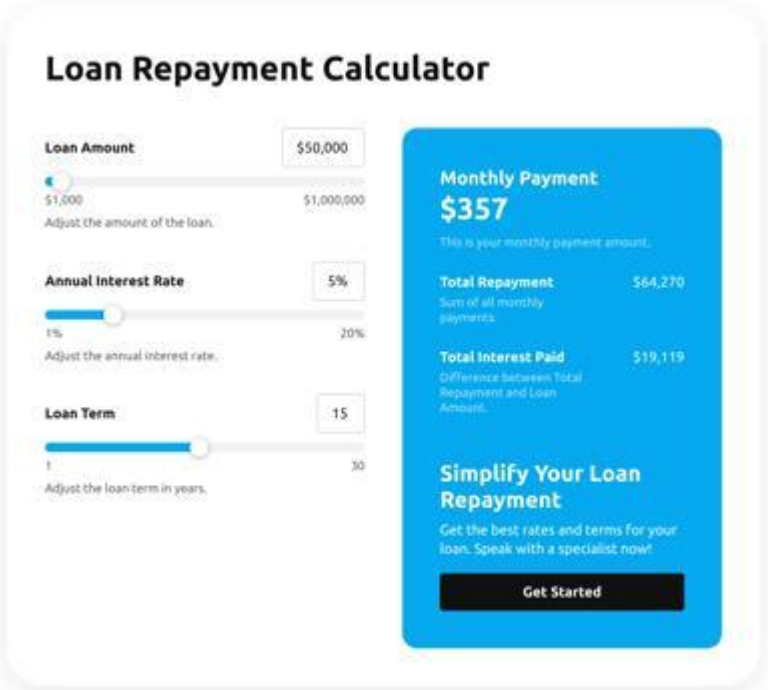
26. Basic Calculator

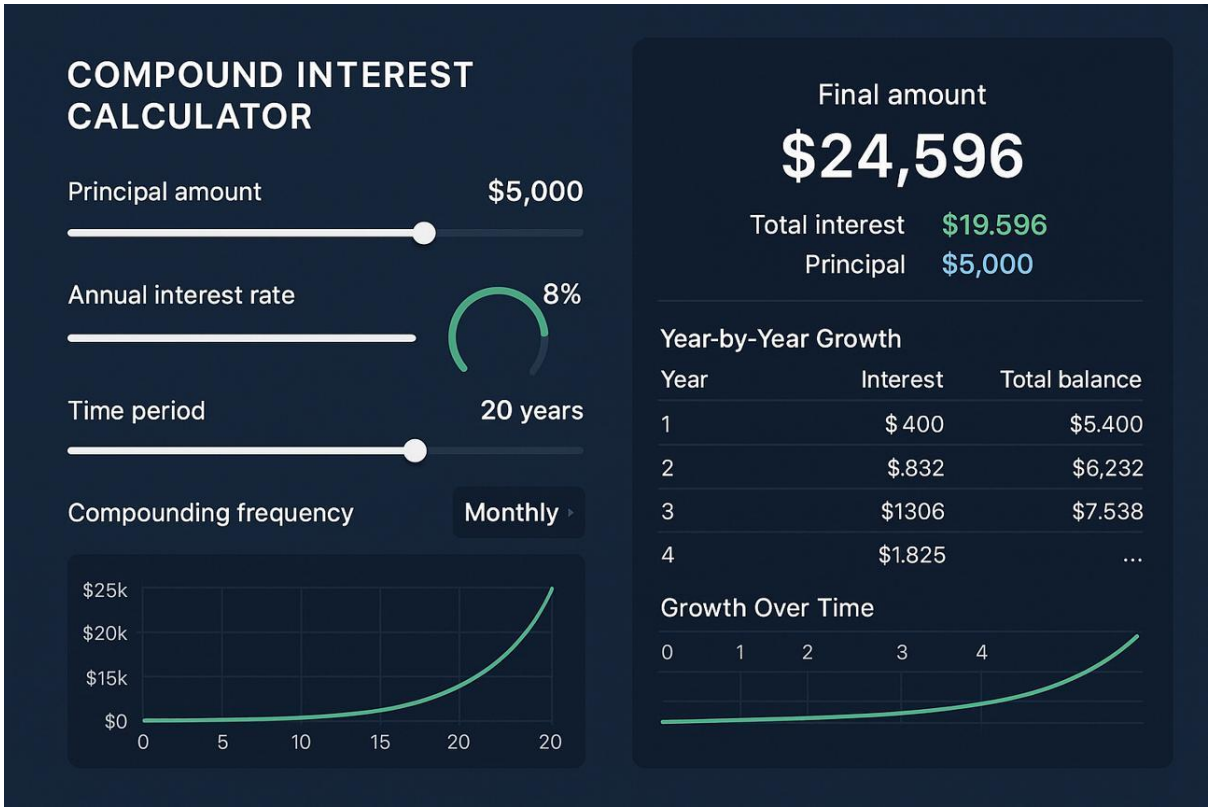
27. Scientific Calculator

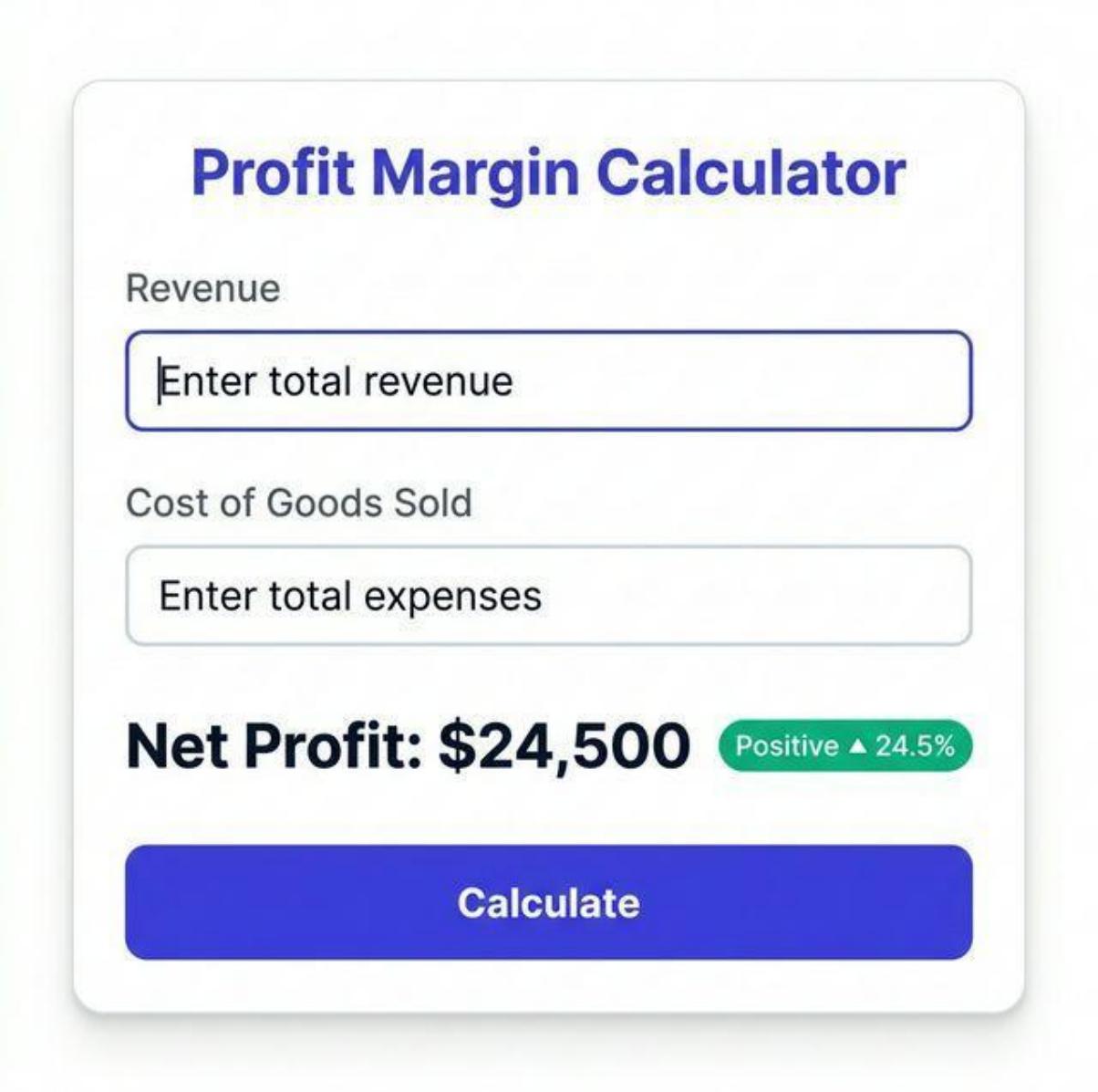
28. Percentage Calculator
29. Percentage Increase/Decrease Calculator
30. Discount Calculator
31. Sales Tax Calculator
32. VAT Calculator
33. Profit Margin Calculator
34. Markup Calculator
35. Ratio Calculator
36. Average Calculator
37. Mean, Median and Mode Calculator
38. Fraction Calculator
39. Exponent Calculator
40. Square Root Calculator
41. LCM Calculator
42. GCD Calculator
43. Prime Number Checker
44. Factorial Calculator
45. Random Number Generator
46. Age Calculator
47. Date Difference Calculator
48. Days Between Dates Calculator
49. Time Duration Calculator
50. Time Zone Converter
51. Percentage Grade Calculator
52. GPA Calculator
53. Standard Deviation Calculator
54. Probability Calculator
55. Matrix Calculator

C. Finance and business calculators

Additional accuracy and trust requirements







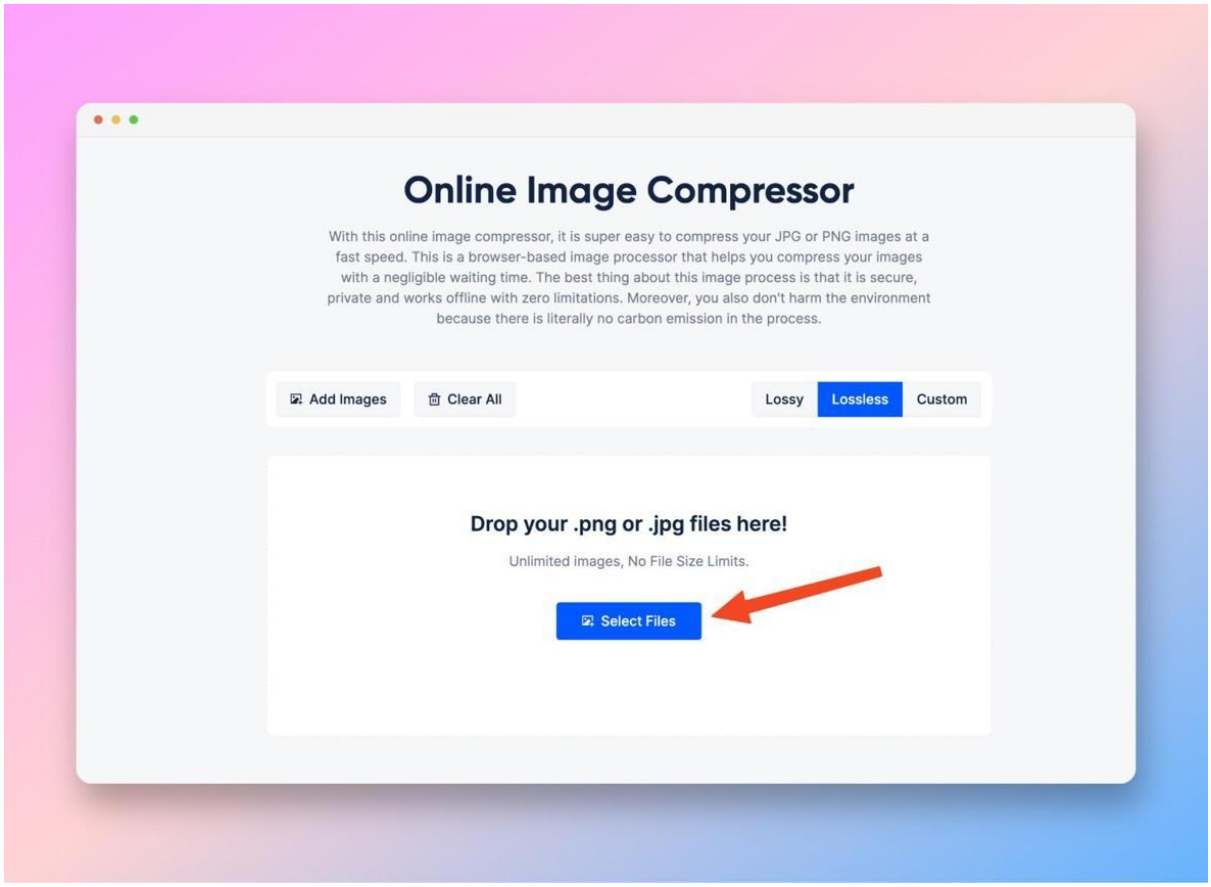
56. Loan Calculator
57. Loan Repayment Calculator
58. Simple Interest Calculator
59. Compound Interest Calculator
60. Savings Calculator
61. Investment Growth Calculator
62. ROI Calculator
63. Break-Even Calculator
64. Business Profit Calculator

65. Business Revenue Calculator
66. Commission Calculator
67. Salary Calculator
68. Hourly to Monthly Salary Converter
69. Overtime Calculator
70. Freelance Rate Calculator
71. Invoice Calculator
72. Budget Calculator
73. Expense Split Calculator
74. Currency Converter
75. Inflation Calculator
76. Mortgage Calculator
77. Debt Payoff Calculator
78. Rental Yield Calculator
79. Stock Average Price Calculator
80. Percentage of Ownership Calculator

For finance-related tools, show assumptions, calculation methods, limitations, and dates for any changing rates. Avoid presenting estimates as guaranteed financial outcomes.

D. Image tools

Strong potential for practical utility traffic



ResizeImage.dev

Install App

Free Image Resizer Online

Resize image directly in your browser. Pick a preset size, adjust a custom crop, and download in JPG, PNG, or WebP without uploading anything.

Drag and Drop Your Image

or click to browse

Select Image

Support: JPEG, JPG, PNG or WebP

Max 50MB per file

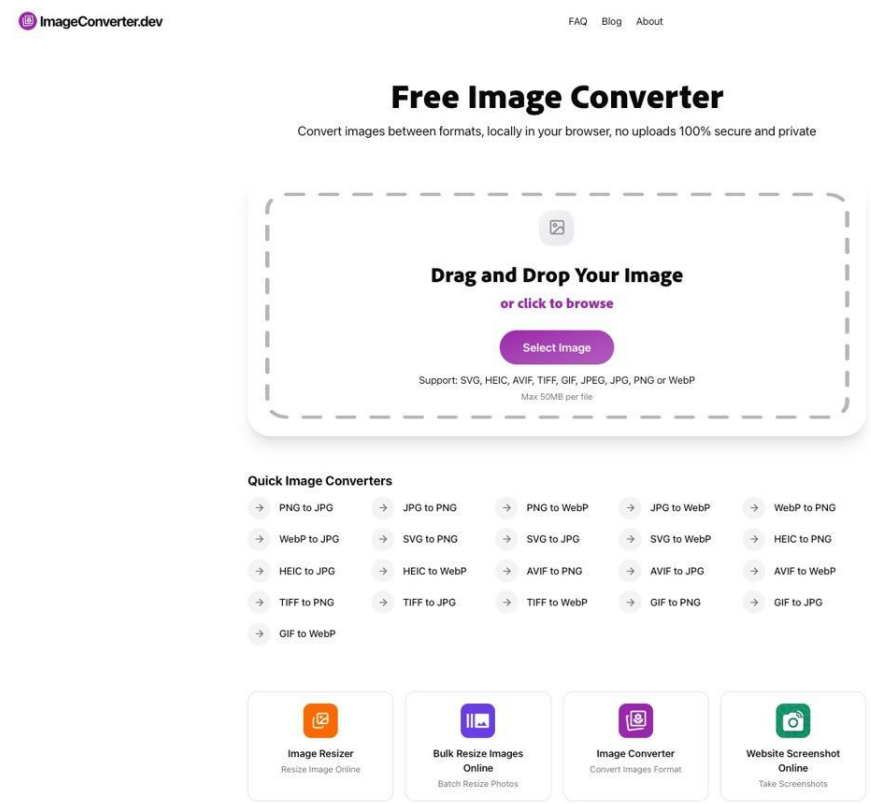
- 100% free image resizer forever
- Lightning-fast
- No sign-up or registration
- Unlimited usage
- Works in browser
- 100% secure & private

Image Resizer
Resize Image Online

Bulk Resize Images Online
Batch Resize Photos

Image Converter
Convert Images Format

Website Screenshot Online
Take Screenshots

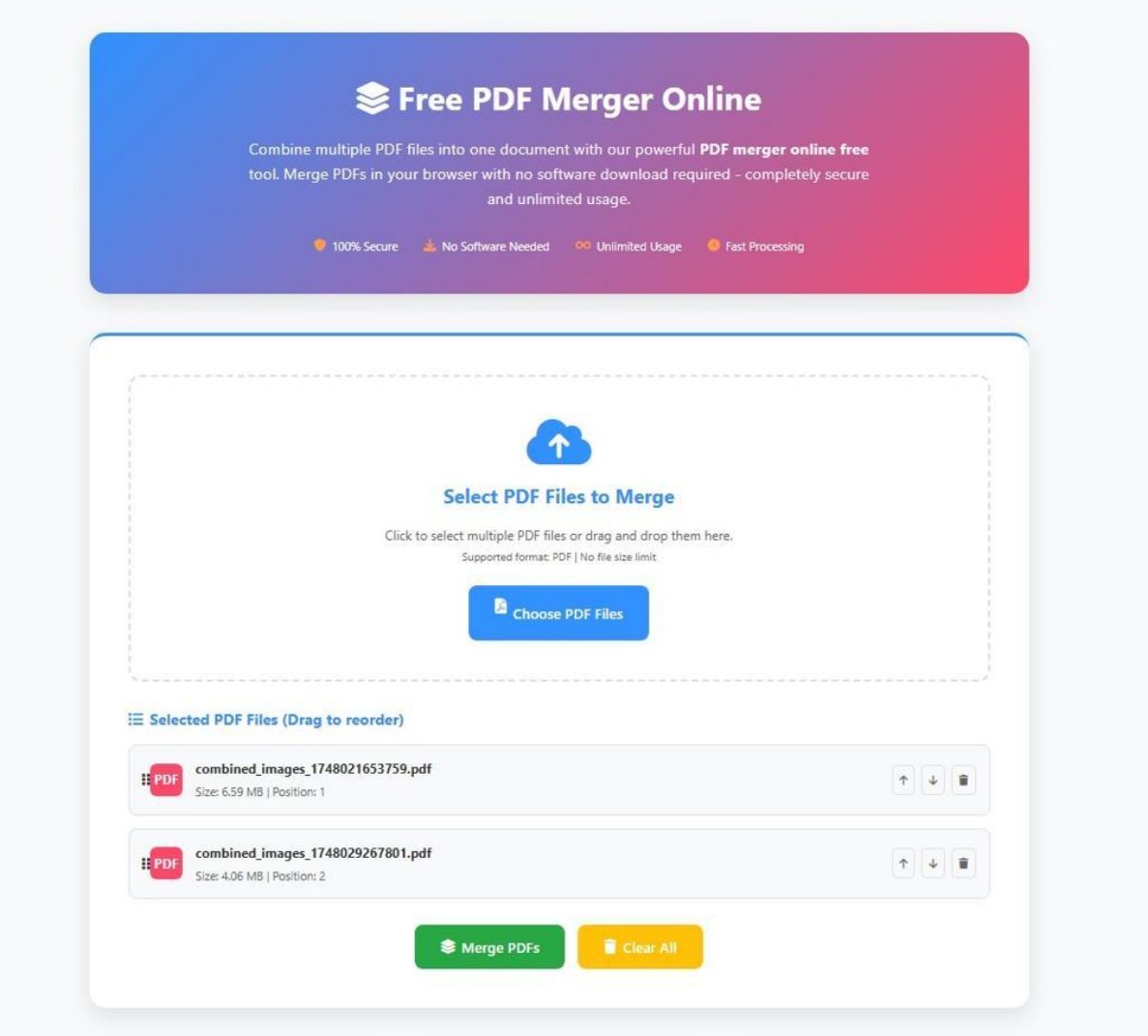


81. Image Compressor
82. Image Resizer
83. Image Cropper
84. Image Format Converter
85. JPG to PNG Converter
86. PNG to JPG Converter
87. WebP Converter
88. Image to Base64 Converter
89. Base64 to Image Converter
90. Image Color Picker
91. Image Resolution Checker
92. Image DPI Calculator
93. Image Metadata Viewer
94. EXIF Data Viewer
95. Favicon Generator

96. Image Blur Tool
97. Image Pixelator
98. Image Background Color Changer
99. Image Dimension Calculator
100. Social Media Image Size Checker

E. PDF and document tools

Strong potential, but more engineering required



Free Online PDF Compressor

Easily reduce the size of your PDF files. Choose your desired compression level for the perfect balance between file size and quality. Fast, free, and all in your browser.

Free & Unlimited · Secure Client-Side · Adjustable Levels · Fast Processing



Drag & Drop PDF Here

or click to select a file from your device

Choose PDF

PDF Details

File Name: The Ultimate Guide to get High PRPATF Tumblr Blogs.pdf

Original Size: 1.05 MB

Pages: 27

Choose Compression Level

Low
(Basic optimization, good quality)

Medium
(Balanced size and quality)

High
(Max reduction, may affect quality)

Note: This tool uses advanced PDF compression combining structural optimization with image compression. For PDFs containing images, actual image re-compression is performed to achieve significant file size reduction. All processing is done securely in your browser - no files are uploaded.

Compress PDF

Clear



101. PDF Merger
102. PDF Splitter
103. PDF Compressor
104. PDF Page Extractor
105. PDF Page Reorder Tool
106. PDF Rotator
107. PDF to JPG Converter
108. JPG to PDF Converter
109. PDF to PNG Converter
110. PNG to PDF Converter
111. PDF Metadata Viewer
112. PDF Page Counter
113. PDF to Text Extractor
114. Text to PDF Converter
115. Word Counter for PDF

116. DOCX to PDF Converter

117. PDF Watermark Tool

118. PDF Password Protection Tool

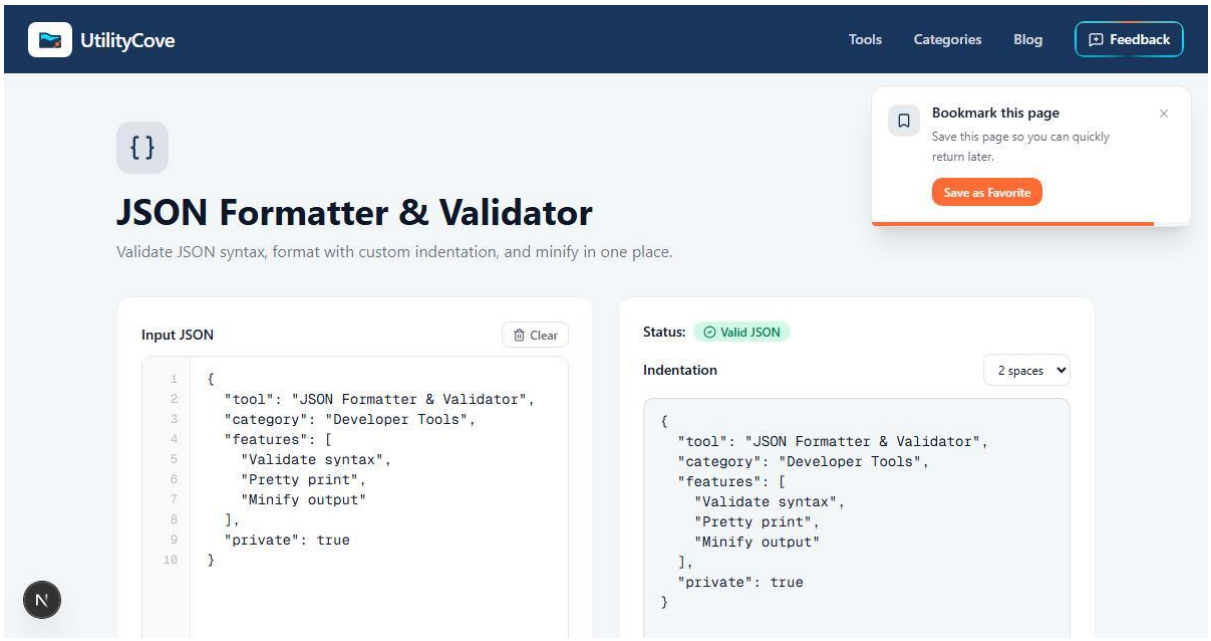
119. PDF File Size Reducer

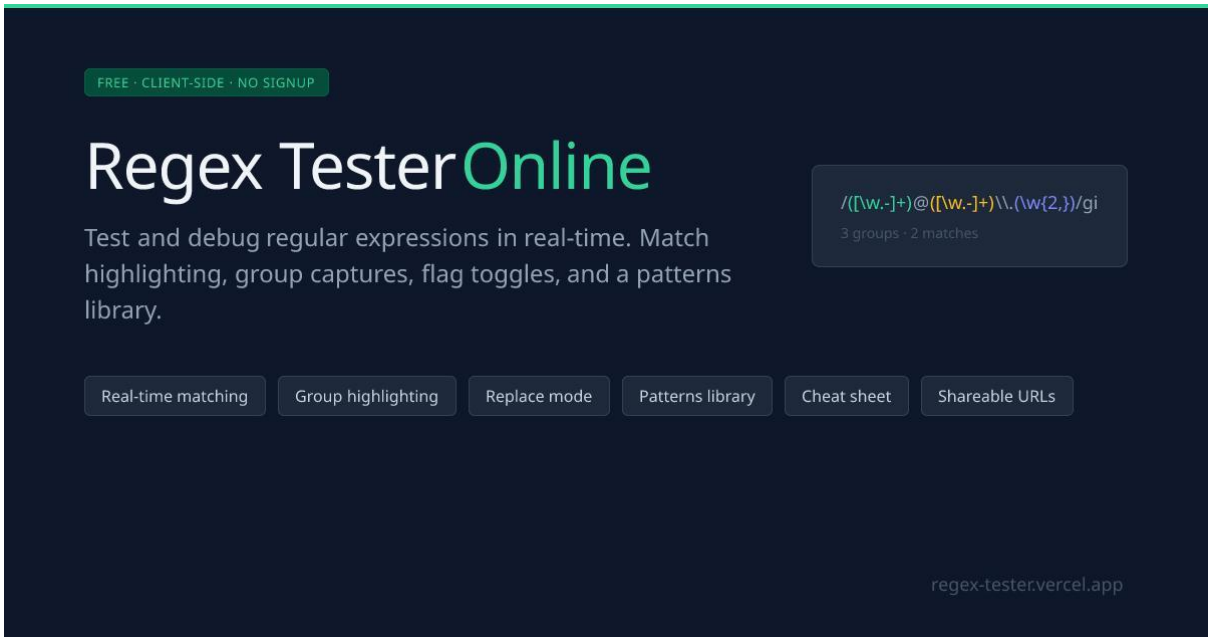
120. Document Word Counter

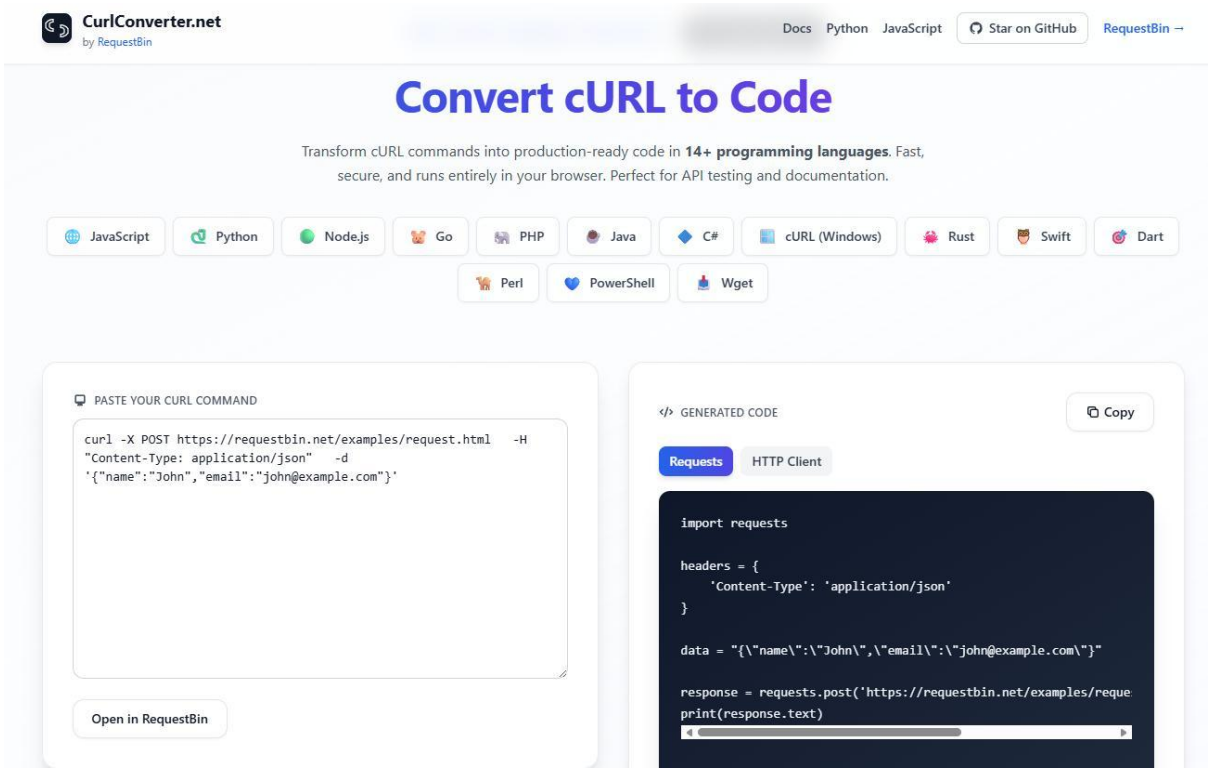
Important: document processing should use secure, temporary file handling. Delete uploaded files automatically and explain how they are processed. Avoid promising that a file is secure or confidential unless your architecture genuinely supports that claim.

F. Developer and programming tools

One of my strongest recommendations







121. JSON Formatter

122. JSON Validator

123. JSON Minifier

124. JSON to CSV Converter

125. CSV to JSON Converter

126. XML Formatter

127. XML Validator
128. HTML Formatter
129. CSS Formatter
130. JavaScript Formatter
131. CSS Minifier
132. JavaScript Minifier
133. HTML Minifier
134. Base64 Encoder
135. Base64 Decoder
136. Hash Generator
137. UUID Generator
138. Password Generator
139. Regex Tester
140. Regex Explainer
141. JWT Decoder
142. Unix Timestamp Converter
143. Unix Timestamp Generator
144. Color Code Converter
145. HEX to RGB Converter
146. RGB to HEX Converter
147. CSS Gradient Generator
148. Box Shadow Generator
149. QR Code Generator
150. QR Code Reader

G. Internet, networking and URL tools

151. URL Shortener
152. URL Parser
153. URL Validator

154. Domain Name Analyzer
155. IP Address Information Tool
156. IPv4 Subnet Calculator
157. DNS Record Lookup
158. HTTP Header Checker
159. User-Agent Parser
160. SSL Certificate Expiry Checker
161. Website Status Checker
162. HTTP Status Code Reference
163. Website Response Time Checker
164. Robots.txt Generator
165. Sitemap XML Generator
166. Meta Tag Generator
167. Open Graph Preview Tool
168. Canonical URL Checker
169. Redirect Checker
170. UTM Link Builder

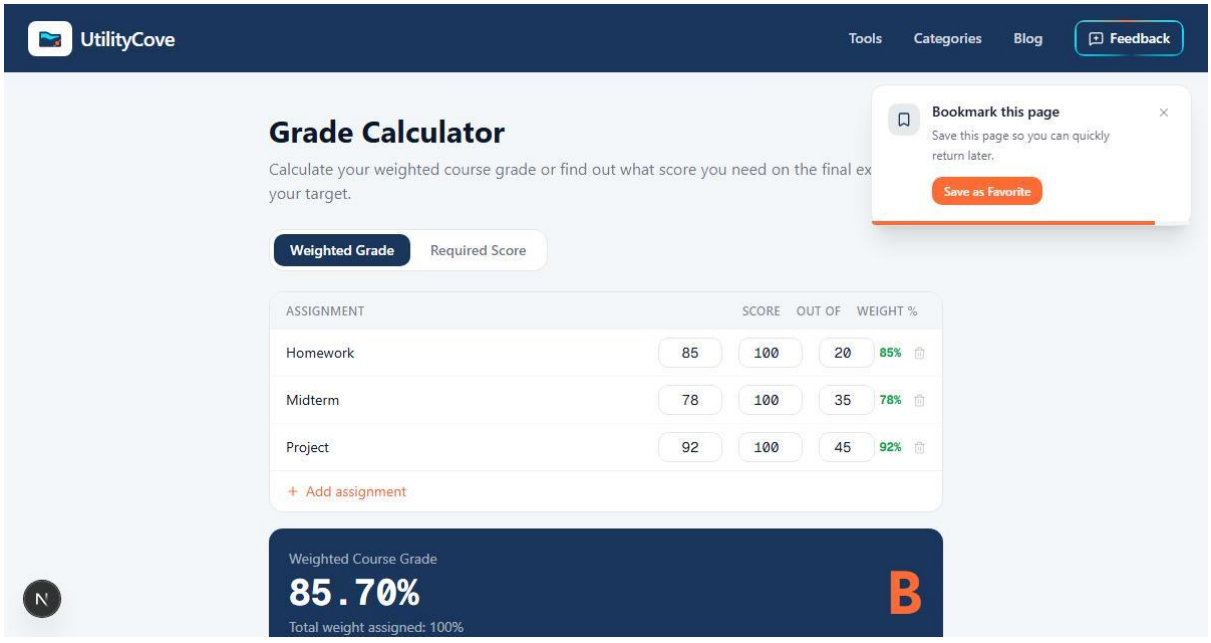
Some of these require external APIs or server-side networking. Never claim to perform a live check when the tool only performs a local simulation.

H. SEO and website management tools

172. Meta Title Previewer
173. Meta Description Previewer
174. SERP Snippet Previewer
175. Keyword Density Analyzer
176. Word Frequency Analyzer
177. SEO Text Analyzer
178. Heading Structure Checker
179. Image Alt Text Checker

180. Internal Link Checker
181. Broken Link Checker
182. Sitemap Validator
183. Robots.txt Tester
184. Schema Markup Generator
185. JSON-LD Validator
186. FAQ Schema Generator
187. Breadcrumb Schema Generator
188. Canonical Tag Generator
189. Hreflang Generator
190. Page Size Analyzer
191. Readability Checker
192. Website Content Outline Generator
193. Keyword Clustering Tool
194. SEO Checklist Generator
195. Website Launch Checklist

I. Education and student tools



Quadratic Equation Solver

Solve $ax^2 + bx + c = 0$ using the quadratic formula

Interactive Solver | Java Code

Enter Coefficients

Enter the values for a, b, and c in the equation $ax^2 + bx + c = 0$

Coefficient a

e.g., 1

Coefficient b

e.g., -5

Coefficient c

e.g., 6

Solve Equation

Quadratic Formula:

$$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$$



196. Grade Calculator
197. GPA Calculator
198. CGPA Calculator
199. Exam Score Calculator
200. Attendance Calculator
201. Study Time Calculator
202. Study Timetable Generator
203. Flashcard Generator
204. Random Quiz Generator
205. Multiplication Table Generator
206. Algebra Calculator
207. Equation Solver
208. Unit Conversion Tool

209. Scientific Notation Converter
210. Roman Numeral Converter
211. Number to Words Converter
212. Word to Number Converter
213. Citation Formatter
214. Bibliography Formatter
215. Essay Word Counter

J. Everyday and lifestyle tools

216. BMI Calculator
217. Calorie Estimator
218. Water Intake Estimator
219. Sleep Duration Calculator
220. Pregnancy Due Date Calculator
221. Ovulation Date Estimator
222. Body Fat Estimator
223. Pace Calculator
224. Running Speed Calculator
225. Cooking Measurement Converter
226. Recipe Scaling Calculator
227. Temperature Converter
228. Weight Converter
229. Length Converter
230. Area Converter
231. Volume Converter
232. Speed Converter
233. Fuel Consumption Converter
234. Electricity Consumption Calculator
235. Countdown Timer

236. Stopwatch
237. Pomodoro Timer
238. World Clock

Health tools require special care: make limitations explicit, avoid diagnosis or treatment claims, and obtain appropriate expert review where necessary.

K. Business, marketing and productivity tools

239. Invoice Generator
240. Receipt Generator
241. Business Name Generator
242. Company Slogan Generator
243. QR Business Card Generator
244. Meeting Cost Calculator
245. Project Cost Calculator
246. Employee Cost Calculator
247. Hourly Rate Calculator
248. Email Signature Generator
249. Business Letter Generator
250. Proposal Template Generator
251. Social Media Caption Formatter
252. Hashtag Generator
253. Content Calendar Generator
254. Marketing Budget Calculator
255. Email Subject Line Analyzer
256. Campaign ROI Calculator
257. Break-Even Point Calculator
258. Product Pricing Calculator
259. Sales Commission Calculator

L. Security and privacy utilities

260. Password Strength Checker
261. Secure Password Generator
262. Random Token Generator
263. Hash Calculator
264. Text Encryption/Decryption Demo
265. File Checksum Calculator
266. SHA-256 Generator
267. SHA-512 Generator
268. URL Safety Syntax Checker
269. Privacy Policy Generator
270. Terms and Conditions Draft Generator
271. Cookie Policy Checklist Generator

Security utilities must be honestly described. A password strength estimator is not a substitute for a professional security audit, and a client-side encryption demo should not claim to provide comprehensive security.

Part 2. Which categories should you build first?

The catalogue contains 271 opportunities, but I would not launch all of them simultaneously.

Here is the recommended order.

Category	**Development difficulty**	**Priority**
Text tools	Very easy	Highest
Mathematics	Easy	Highest
Developer tools	Easy–medium	Highest
Everyday converters	Easy	High
SEO utilities	Easy–medium	High

Education	Easy–medium	High
Business calculators	Medium	High
Image tools	Medium	High
PDF tools	Medium–hard	Medium
Internet tools	Medium–hard	Medium
Marketing tools	Medium	Medium
Health tools	Medium–hard	Later
Security tools	Medium–hard	Later

My recommended initial launch: 30–40 fully functional tools across four categories:

1. Text and writing.
2. Mathematics and calculators.
3. Developer utilities.
4. Everyday converters.

This is a more manageable launch than 271 unfinished tools.

Part 3. What Google AdSense actually requires

This is the most important part of the project.

Google does not maintain an official list of web-tool categories that automatically qualify for AdSense. A word counter, calculator, image converter or PDF tool is not automatically approved simply because it is useful.

Google evaluates the website and its content, not just the names of its tools.

Here are the requirements we need to engineer around.

3.1 Original and valuable content

Google's eligibility guidance emphasizes:

- Original, interesting content.
- Content that is relevant to visitors.

- A good user experience.
- Clear navigation.
- Ownership and access to the website's HTML source.
- Compliance with the AdSense Program Policies.

Source: AdSense eligibility requirements and Make sure your site's pages are ready for AdSense.

How we implement this:

Every tool should have its own properly developed page.

For example, a percentage calculator should not simply contain a title, an input box and a result.

It should contain:

1. A working percentage calculator.
2. A clear explanation of what it calculates.
3. Step-by-step calculation methodology.
4. At least three relevant worked examples.
5. Common use cases.
6. Input validation and error handling.
7. Frequently asked questions.
8. Related tools.
9. Relevant limitations or disclaimers.
10. A useful mobile-friendly interface.

The content must be genuinely useful. Do not generate identical paragraphs and replace the tool name for hundreds of pages.

Google specifically identifies mass-produced, unoriginal pages with little user value as a form of scaled content abuse.



Google for Developers+1

3.2 Essential website pages

I recommend creating these pages before applying.

Page	Purpose	Priority
Homepage	Explains the platform and showcases tools	Essential
About Us	Identifies the publisher and mission	Essential
Contact Us	Provides a working contact channel	Essential
Privacy Policy	Explains data collection and advertising	Essential
Terms of Service	Defines how visitors may use the platform	Strongly recommended
Disclaimer	Explains limitations of calculations and outputs	Strongly recommended
Cookie Policy	Explains cookie and tracking practices	Strongly recommended
Affiliate Disclosure	Discloses affiliate relationships, if any	Conditional
Editorial Policy	Explains how guides are researched and reviewed	Recommended
Corrections Policy	Explains how errors are reported and corrected	Recommended
Accessibility Statement	Explains accessibility commitments	Recommended
Copyright/DMCA Policy	Provides a copyright complaint process	Recommended

These are recommendations for a robust, trustworthy platform. Do not mistake every suggested page for an independently mandated AdSense application requirement.

What each important page must contain

About Us

- Who operates the website.
- What the website provides.
- Why the tools exist.
- Who the tools are intended to help.
- How tools are developed and maintained.
- Real contact or publisher information.

Privacy Policy

It should accurately disclose:

- What information is collected.
- Whether you use analytics.
- Whether cookies and local storage are used.
- Whether uploaded files are processed.
- Whether information is shared with service providers.
- How Google advertising works on your website.
- How third-party vendors may use cookies, web beacons, IP addresses or other identifiers.
- How users can manage relevant privacy choices.
- How users can contact you about privacy.

Google explicitly requires appropriate privacy disclosures when using its advertising services.



Google AdSense Help+1

Do not copy a random website's privacy policy without adjusting it to your actual technology.

3.3 Ad placement requirements

This is where many otherwise useful tools websites can create problems.

Google prohibits deceptive ad placements and implementations that encourage accidental clicks.

Avoid:

- Ads immediately beside download buttons.
- Ads that look like calculator buttons.
- Ads disguised as navigation links.
- Ads placed where visitors are likely to click accidentally.
- Arrows pointing toward advertisements.
- Text encouraging people to click ads.
- Pop-under advertisements.
- Automatically refreshing ads without a qualifying user request.
- Ads covering the main functionality.
- Ads on dead-end or empty screens.

Google AdSense Help+2

Recommended page layout

YOUR BRAND

Logo · Search tools · Categories · About

Percentage Calculator

Calculate percentages, increases and decreases.

Enter your number

250

Enter percentage

20

Calculate

Result: 50

Illustrative ad placement

Advertisement

Distinct from the calculator and its controls.

How to use this calculator

Explain the formula, demonstrate examples, explain common mistakes and provide related information.

Advertisement – optional additional placement

Frequently asked questions

- How is the percentage calculated?
- Can I calculate percentage increases?
- How accurate is this result?

Footer: About · Contact · Privacy · Terms · Disclaimer · Cookie Settings

Conceptual page structure, not a Google-approved placement template. Actual ad eligibility and placement depend on the page and implementation.

My recommendation is to keep advertisements separate from the calculator's interactive area, especially from its input, calculate, copy, reset and download controls.

3.4 Avoid low-value tool pages

Google has a specific policy concerning screens without publisher content or with low-value content.

A page that contains only an input box, an output box and advertisements can be problematic.

Likewise, error pages, unfinished pages and empty pages should not carry ads.



Publisher Policies Help+1

Our solution:

- Make each tool functional.
- Include meaningful explanations.
- Include real examples.
- Explain limitations.
- Provide related information.
- Test the results.
- Do not place ads on empty or error screens.

3.5 Traffic quality

Do not buy suspicious traffic or artificially generate visits.

Never:

- Click your own ads.
- Ask friends to click advertisements.
- Pay people to click ads.
- Use bots to generate page views.
- Use automated clicking software.
- Participate in click-exchange schemes.
- Use misleading traffic sources.

Google identifies artificial clicks and impressions as invalid traffic and can restrict or terminate monetization.



Google AdSense Help+1

Build traffic through useful products, search visibility, genuine referrals and legitimate marketing.

3.6 Privacy and consent

This requires special attention if you expect visitors from different countries.

For users in the European Economic Area, the United Kingdom and Switzerland, Google requires a Google-certified CMP integrated with the IAB Transparency and Consent Framework when serving personalized ads.

Google provides its own consent-management options and maintains a list of certified providers.



Google AdSense Help+1

Our implementation should:

- Use an appropriate consent management solution.
- Provide understandable privacy disclosures.
- Respect user choices.
- Ensure advertising tags follow the applicable consent signals.

- Avoid loading nonessential tracking before the required consent.
- Allow users to revisit their privacy choices.

Do not assume that a generic cookie popup is sufficient.

3.7 Site verification and application

Google's site approval process involves reviewing the submitted website. Google says the process often takes a few days but can take two to four weeks.

You need to connect the site properly and complete account and payment information requirements.



Google AdSense Help+1

Before applying:

- Verify domain ownership.
- Ensure your website loads correctly.
- Make the site accessible to Google's crawler.
- Make sure important pages are publicly accessible.
- Avoid blocking necessary Google crawlers.
- Add the AdSense code as instructed.
- Check the site status inside your AdSense account.
- Fix any issues reported during review.

Official reference: Connect your site to AdSense.

Part 4. The technical architecture

Now we move from policy to engineering.

I recommend using a modern framework rather than building 271 separate websites or separate applications.

4.1 Recommended technology stack

Component	Recommended technology	Reason
Frontend	Next.js	SEO-friendly web application

Language	TypeScript	Fewer development errors
Styling	Tailwind CSS	Consistent responsive design
UI components	shadcn/ui	Reusable interface components
Icons	Lucide	Consistent icons
Form validation	Zod	Input validation
Calculations	JavaScript/TypeScript	Fast local processing
Backend	Next.js server functions/API routes	Server-side functionality
Database	PostgreSQL	Structured data
Database hosting	Supabase	Managed PostgreSQL option
File processing	Browser APIs initially	Reduced server costs
Image processing	Canvas API / browser libraries	Client-side image operations
PDF processing	PDF.js and appropriate PDF libraries	Document processing
Hosting	Vercel or equivalent	Deployment and hosting
Analytics	Google Analytics 4	Traffic measurement
Search monitoring	Google Search Console	Indexing and search performance
Advertising	Google AdSense	Monetization

You do not need all these services immediately.

For the initial launch, a static or server-rendered application with local calculations is sufficient for many tools.

4.2 The website architecture

HomepageTool DirectoryCategoriesIndividual Tool PagesInteractive Tool EngineOriginal Explanatory ContentRelated ToolsClient-side ProcessingBackend APIs when requiredDatabase or Temporary StorageAdSense IntegrationAbout / Contact / PoliciesGoogle Search ConsoleAnalyticsConsent Management

4.3 Reusable tool architecture

Rather than coding every calculator from scratch, create a shared component system.

For example:

```
src/
├── app/
│   ├── page.tsx
│   ├── tools/
│   │   ├── page.tsx
│   │   ├── word-counter/
│   │   │   └── page.tsx
│   │   ├── percentage-calculator/
│   │   │   └── page.tsx
│   │   └── json-formatter/
│   │       └── page.tsx
│   ├── categories/
│   ├── about/
│   ├── contact/
│   ├── privacy-policy/
│   ├── terms/
│   └── disclaimer/
├── components/
```

```
│   ├── Header.tsx
│   ├── Footer.tsx
│   ├── ToolCard.tsx
│   ├── ToolSearch.tsx
│   ├── CalculatorInput.tsx
│   ├── ResultPanel.tsx
│   ├── CopyButton.tsx
│   ├── AdSlot.tsx
│   └── ConsentManager.tsx
├── lib/
│   ├── calculations/
│   ├── validators/
│   ├── formatters/
│   ├── analytics/
│   └── security/
├── data/
│   ├── tools.ts
│   └── categories.ts
└── styles/
```

This structure lets us add new tools without rebuilding the entire website.

4.4 Tool data model

Every tool should have a consistent definition.

For example:

```
type ToolDefinition = {
  id: string;
  name: string;
  slug: string;
```

```
 category: string;
 description: string;
 keywords: string[];
 difficulty: "easy" | "medium" | "advanced";
 processing: "client" | "server";
 hasDownload: boolean;
 hasUpload: boolean;
 seoTitle: string;
 seoDescription: string;
 formula?: string;
 content: {
  introduction: string;
  instructions: string[];
  examples: string[];
  limitations: string[];
  faqs: {
   question: string;
   answer: string;
  }[];
 };
};
```

This creates consistency across the entire platform.

However, the content must still be individually researched and written. A reusable template should not turn into a mass-production system for low-value pages.

4.5 UI/UX standards

Every tool should have:

- Responsive mobile-first layout.
- Clear tool title.

- Short description.
- Obvious input fields.
- Clear primary action.
- Immediate results where appropriate.
- Copy and reset functionality.
- Input validation.
- Accessible labels.
- Keyboard navigation.
- Helpful error messages.
- Loading indicators for longer processes.
- Relevant examples.
- Related tools.
- Privacy information when files or sensitive data are processed.

The design should feel like a professional productivity platform, not a collection of SEO landing pages.

Part 5. How to implement each tool category

The following is the engineering plan. It explains what each tool actually does, the technology required, the main development tasks, and what makes the resulting page valuable.

5.1 Text and writing tools

Phase 1

Tool	**Main functionality**	**Implementation**
Word Counter	Words, characters, sentences, paragraphs	Text parsing
Case Converter	Multiple letter-case transformations	String functions
Duplicate Remover	Remove repeated lines	Set and array processing

Text Difference Checker	Highlight text differences	Diff algorithm
Reading Time	Estimate reading duration	Word count / reading speed
Speaking Time	Estimate speech duration	Word count / speaking speed
Text Reverser	Reverse characters or words	String manipulation
Alphabetical Sorter	Sort lines A–Z or Z–A	Array sorting
Extra Space Remover	Normalize spaces	Regular expressions
Slug Converter	Generate URL-friendly slugs	Unicode normalization
Find and Replace	Search and replace text	String operations
Markdown Previewer	Render Markdown safely	Markdown parser
HTML Encoder	Encode special characters	HTML entity functions
URL Encoder	Encode and decode URLs	URL APIs
Lorem Ipsum Generator	Generate placeholder text	Local text generator

Development specification

The text-processing engine should:

1. Accept multiline text.
2. Handle Unicode characters.
3. Support large inputs without freezing the browser.
4. Display results immediately.

5. Provide a copy button.
6. Provide a clear/reset button.
7. Handle empty inputs.
8. Avoid uploading text unnecessarily.

Important: A text tool can be fully functional without a backend. This reduces cost, latency and unnecessary data collection.

5.2 Calculator tools

Phase 1

Tool	**Required calculation**	**Main engineering requirement**
Basic Calculator	Arithmetic	Expression parser
Scientific Calculator	Advanced mathematics	Math engine
Percentage Calculator	Part of a whole	Formula validation
Discount Calculator	Discounted price	Currency formatting
VAT Calculator	Tax-inclusive/exclusive values	Configurable tax rate
Profit Margin	Profit / revenue	Zero-denominator protection
Ratio Calculator	Equivalent ratios	GCD simplification
Average Calculator	Arithmetic mean	Numeric validation
Age Calculator	Age between dates	Calendar calculations
Date Difference	Days between dates	Date handling
Time Duration	Difference between times	Time-zone awareness

GPA Calculator	Weighted academic grades	Configurable grading scale
Standard Deviation	Statistical variation	Population/sample modes
Probability	Event probability	Clearly defined assumptions
Matrix Calculator	Matrix operations	Dimension validation

Every calculator should have its formula documented and tested against known results.

For instance:

Percentage calculation

$$\text{Result} = \frac{\text{Percentage}}{100} \times \text{Number}$$

Example:

$$20\% \times 250 = 50$$

Include alternative calculations, such as percentage increase and percentage decrease, as separate modes rather than confusing them with the basic percentage function.

5.3 Finance and business tools

Phase 2 – Expert review recommended

These tools require greater attention to numerical accuracy.

Tool	**Main functionality**	**Important considerations**
Loan Calculator	Repayment estimation	Interest and payment frequency
Compound Interest	Investment growth	Compounding interval
ROI Calculator	Return on investment	Investment and return definitions

Savings Calculator	Savings projection	Contributions and rates
Salary Calculator	Pay estimation	Pay period and deductions
Budget Calculator	Income and expenses	Currency and categories
Debt Calculator	Debt repayment projection	Rates and payment assumptions
Break-Even Calculator	Revenue needed to cover costs	Fixed and variable costs
Rental Yield	Property return estimation	Gross versus net yield
Currency Converter	Exchange conversion	Freshness of exchange rates

For these pages, include:

- Formula explanations.
- Worked examples.
- Assumptions.
- Limitations.
- Currency and regional settings where relevant.
- Source and update time for external financial data.

Do not imply that an estimate represents an official tax, lending, investment or financial decision.

5.4 Image tools

Phase 2

Tool	**Main functionality**	**Suggested technology**
Image Compressor	Reduce file size	Browser image processing
Image Resizer	Change dimensions	Canvas API

Image Cropper	Crop an image	Canvas API
JPG to PNG	Format conversion	Canvas API
PNG to JPG	Format conversion	Canvas API
WebP Converter	Convert to WebP	Browser image encoding
Base64 Converter	Encode image data	FileReader
Color Picker	Extract image colors	Canvas pixel sampling
EXIF Viewer	Read image metadata	Metadata parsing library
Favicon Generator	Create website icons	Canvas and file export
Image Pixelator	Apply pixelation	Canvas processing
Image Blur	Blur an image	Canvas filters

Recommended processing approach:

Visitor selects an imageValidate file type and sizeLoad image locallyProcess using browser APIsPreview outputAllow downloadRelease temporary resources

For the initial version, prioritize browser-side processing. Do not store uploaded images on your server unless a feature genuinely requires it.

5.5 PDF tools

Phase 3 – More complex

Tool	**Main functionality**	**Technology**
PDF Merger	Combine PDFs	PDF-lib
PDF Splitter	Extract page ranges	PDF-lib
PDF Compressor	Reduce PDF size	Specialized PDF processing

PDF Rotator	Rotate pages	PDF-lib
PDF Page Extractor	Select pages	PDF-lib
PDF to JPG	Render PDF pages	PDF.js
JPG to PDF	Create PDF	PDF-lib
PDF Text Extractor	Extract text	PDF.js
PDF Watermark	Apply text/image watermark	PDF-lib
PDF Metadata Viewer	Inspect document properties	PDF metadata libraries

For PDF tools, build a common upload and processing engine.

Required safeguards:

- Validate file signatures, not only extensions.
- Set maximum upload size.
- Set processing time limits.
- Reject malformed documents.
- Protect against malicious file uploads.
- Avoid exposing file paths.
- Use temporary storage.
- Automatically delete server-side temporary files.
- Explain file-processing practices.
- Test with large and unusual PDFs.

5.6 Developer tools

Phase 1

Tool	**Implementation**
JSON Formatter	JSON parser and serializer

JSON Validator	JSON.parse with error reporting
JSON Minifier	JSON serialization
CSV to JSON	CSV parsing
JSON to CSV	Structured conversion
XML Formatter	XML parser and serializer
HTML Formatter	HTML parser
CSS Formatter	CSS parser/formatter
JavaScript Formatter	JavaScript formatter
Base64 Encoder	Browser Base64 APIs
UUID Generator	Cryptographically secure UUID API
Password Generator	Cryptographic random number generator
Regex Tester	Regular expressions
JWT Decoder	Base64URL decoding
Timestamp Converter	Date and time conversion
Color Converter	Color transformations
CSS Gradient Generator	CSS generation
Box Shadow Generator	CSS generation
QR Generator	QR encoding library

QR Reader	Browser camera/image processing

Security requirements:

- Never execute arbitrary submitted JavaScript on your main website origin.
- Do not send secrets to third-party APIs unnecessarily.
- Do not log passwords, tokens or private input.
- Do not store submitted developer code without a clear reason and disclosure.
- Treat uploaded files and pasted code as untrusted input.

5.7 SEO tools

These tools need special attention because they can attract website owners, developers and digital marketers.

Tool	**Functionality**
Meta Title Previewer	Preview search titles
Meta Description Previewer	Preview descriptions
SERP Previewer	Approximate search-result appearance
Keyword Density	Calculate word frequencies
Heading Checker	Inspect heading hierarchy
Alt Text Checker	Identify images without alt text
Sitemap Generator	Create XML sitemaps
Robots.txt Generator	Generate robots rules
Schema Generator	Create structured data
Canonical Generator	Generate canonical tags

Hreflang Generator	Create language-region annotations
Open Graph Preview	Preview social metadata
Redirect Checker	Inspect redirect behavior
UTM Builder	Create tracking URLs
Readability Checker	Estimate text readability

Some of these can operate entirely in the browser. Others, such as live website inspection, need a carefully secured backend because browser cross-origin restrictions prevent arbitrary website requests.

Do not use your server as an unrestricted URL-fetching proxy. That could create server-side request forgery vulnerabilities.

Part 6. The standard blueprint every individual tool must follow

This is the most important reusable implementation specification.

Every tool will follow the same development lifecycle.

Stage 1: Product research

The product analyst must determine:

- What problem the tool solves.
- Who needs the tool.
- How users currently solve that problem.
- What competitors offer.
- What is missing from existing solutions.
- What makes our tool more useful.
- What functionality must be included in the first release.

Deliverable: A one-page product requirements document.

Stage 2: UI/UX design

The designer creates:

1. Desktop layout.

2. Mobile layout.
3. Tablet layout.
4. Input states.
5. Output states.
6. Error states.
7. Loading states.
8. Empty states.
9. Success states.
10. Accessibility specifications.

The interface must make the primary task immediately understandable.

Stage 3: Frontend development

The developer implements:

- User inputs.
- Validation.
- Calculation or processing.
- Results.
- Copy functionality.
- Download functionality where applicable.
- Reset functionality.
- Responsive behavior.
- Accessibility.

Stage 4: Backend development

A backend should only be introduced when needed.

Examples:

- Live exchange-rate data.
- Domain and DNS checks.
- Server-side PDF processing.
- File conversion requiring server resources.

- User accounts.
- Saved projects.
- Usage history.

A simple percentage calculator does not need a database or a backend.

Stage 5: Content development

Each tool page needs an original editorial package.

My recommended editorial structure:

Content section	**What it should achieve**
Introduction	Explain the problem
Tool instructions	Explain how to use it
Formula or method	Show how results are produced
Worked examples	Demonstrate actual usage
Practical applications	Explain real-world uses
Accuracy and limitations	Set appropriate expectations
Frequently asked questions	Resolve common uncertainties
Related tools	Help users discover useful functionality

There is no official AdSense minimum word count for each tool. We should not manufacture unnecessary paragraphs simply to reach an arbitrary word target.

Stage 6: SEO development

Each page receives:

- Unique title tag.
- Unique meta description.
- Descriptive URL.
- One clear primary heading.

- Logical heading hierarchy.
- Descriptive internal links.
- Canonical URL.
- Relevant structured data where appropriate.
- Optimized images.
- Search-engine-accessible content.
- Appropriate indexing settings.

Example:

/tools/percentage-calculator/

Example page title:

Percentage Calculator – Calculate Percentages Easily

Example meta description:

Calculate percentages, percentage increases and decreases.

Use free examples and instant results with our percentage calculator.

Do not create hundreds of pages targeting nearly identical keywords with no meaningful difference.

Google's people-first content guidance and spam policies are particularly relevant to this strategy.



Google for Developers+1

Stage 7: Quality assurance

Every tool must pass the following tests.

Test	**Acceptance criteria**
Functional	Produces correct results
Validation	Rejects invalid inputs
Mobile	Works on small screens

Desktop	Works at common desktop sizes
Accessibility	Keyboard and screen-reader support
Performance	No unnecessary processing delays
Security	No obvious input-handling vulnerabilities
Privacy	Does not collect unnecessary information
SEO	Proper metadata and indexability
Advertising	No misleading or accidental-click placements

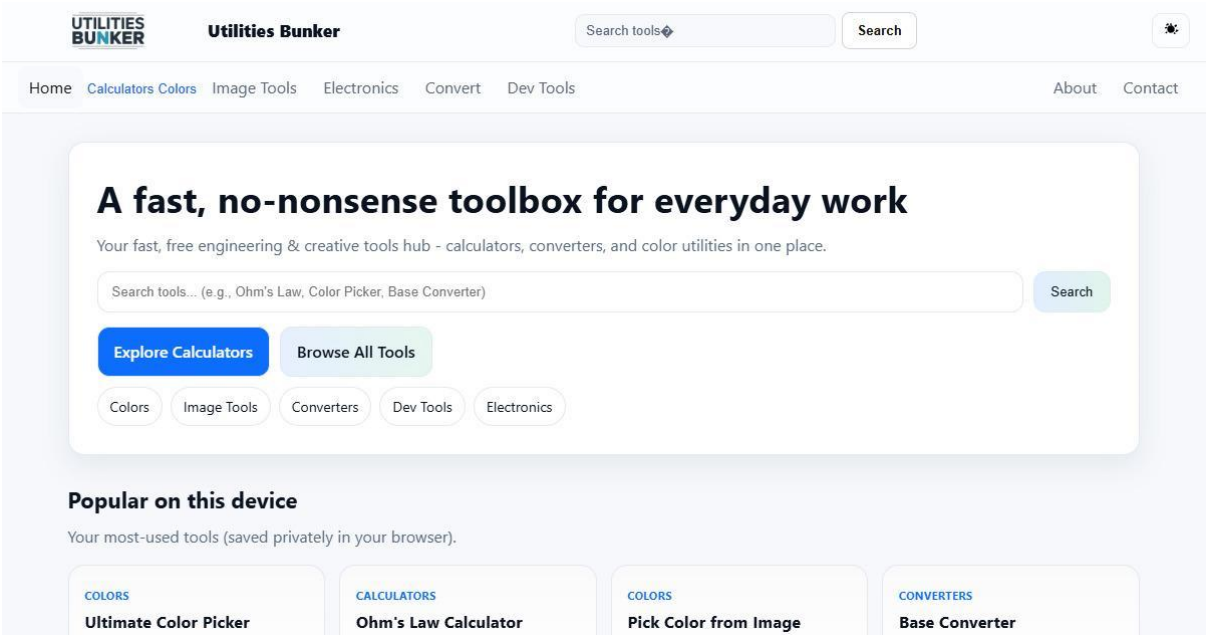
Stage 8: Launch

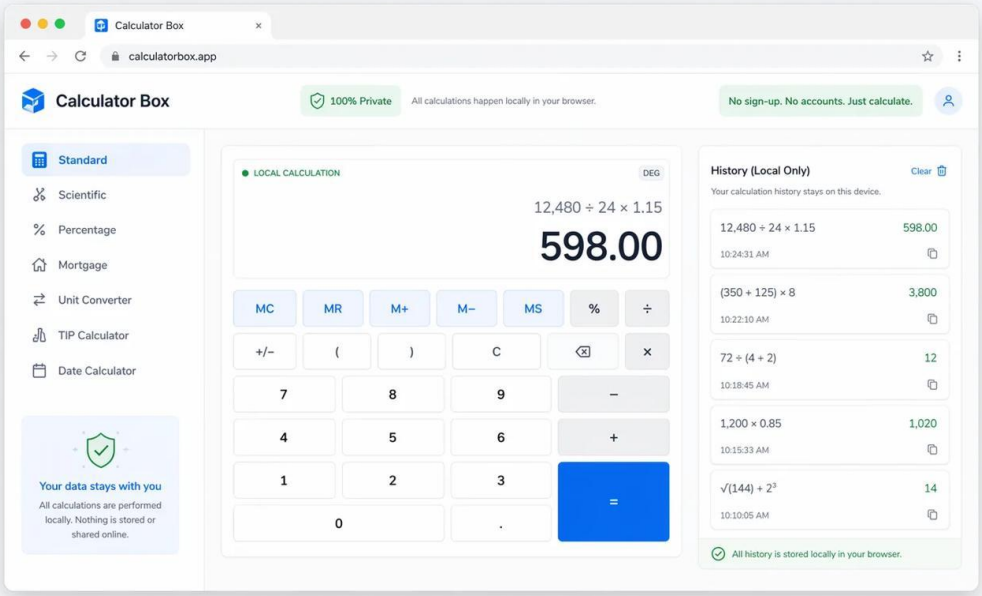
Only after testing:

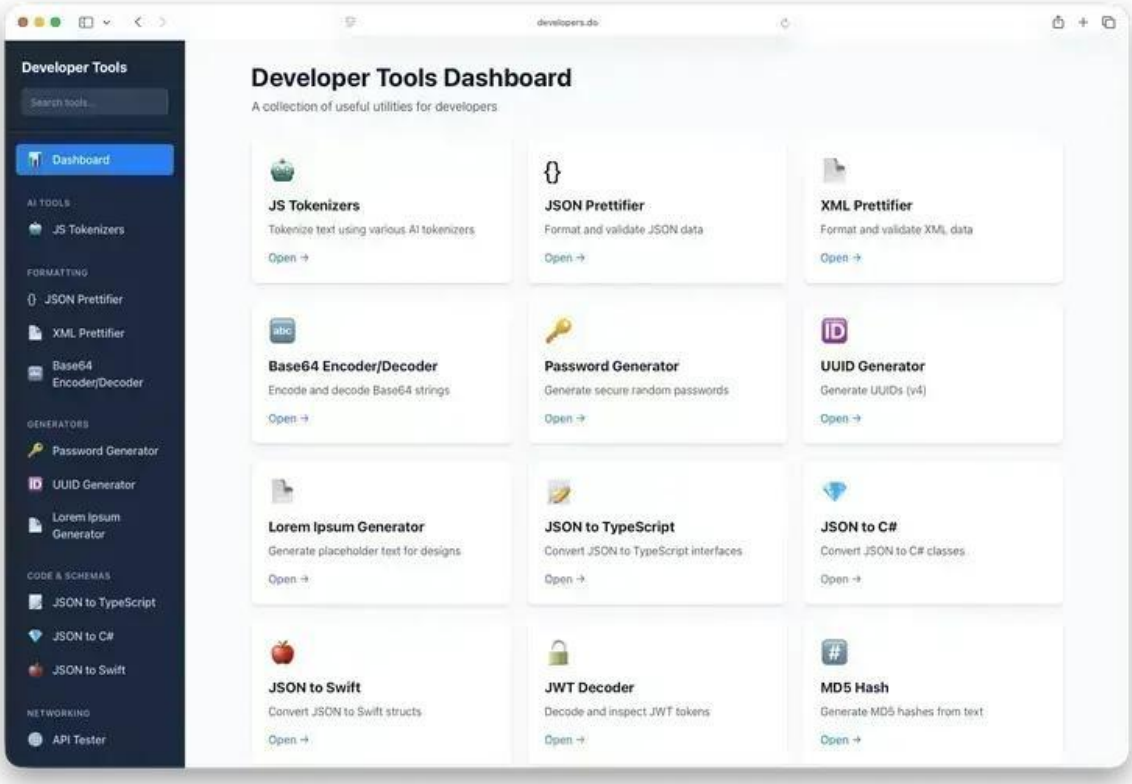
- Publish the page.
- Add it to the category directory.
- Add relevant internal links.
- Update the sitemap.
- Submit or notify Search Console as appropriate.
- Monitor indexing and errors.
- Monitor user feedback.
- Fix defects.

Part 7. Website design and user experience

The entire platform should use one consistent visual system.


UTILITIES BUNKER
Utilities Bunker
Search tools
Search
Home
Calculators Colors
Image Tools
Electronics
Convert
Dev Tools
About
Contact
A fast, no-nonsense toolbox for everyday work
Your fast, free engineering & creative tools hub - calculators, converters, and color utilities in one place.
Search tools... (e.g., Ohm's Law, Color Picker, Base Converter)
Search
Explore Calculators
Browse All Tools
Colors
Image Tools
Converters
Dev Tools
Electronics
Popular on this device
Your most-used tools (saved privately in your browser).
COLORS
Ultimate Color Picker
CALCULATORS
Ohm's Law Calculator
COLORS
Pick Color from Image
CONVERTERS
Base Converter


Calculator Box
calculatorbox.app
Calculator Box
100% Private
All calculations happen locally in your browser.
No sign-up. No accounts. Just calculate.
Standard
Scientific
Percentage
Mortgage
Unit Converter
TIP Calculator
Date Calculator
Your data stays with you
All calculations are performed locally. Nothing is stored or shared online.
LOCAL CALCULATION
DEG
12,480 ÷ 24 × 1.15
598.00
MC
MR
M+
M−
MS
%
÷
+/−
(
)
C
×
7
8
9
−
4
5
6
+
1
2
3
=
0
.
History (Local Only)
Clear
Your calculation history stays on this device.
12,480 ÷ 24 × 1.15
598.00
10:24:31 AM
(350 + 125) × 8
3,800
10:22:10 AM
72 ÷ (4 + 2)
12
10:18:45 AM
1,200 × 0.85
1,020
10:15:33 AM
√(144) + 2³
14
10:10:05 AM
All history is stored locally in your browser.



7.1 Homepage structure

Recommended order:

1. Header and logo.
2. Main navigation.
3. Hero section explaining the website.
4. Prominent tool search.
5. Popular tools.
6. Main tool categories.
7. Recently added tools.
8. How the platform works.
9. Why visitors should use it.
10. Helpful guides.
11. Frequently asked questions.
12. Footer and policy links.

7.2 Category pages

Each category should have:

- Category introduction.
- Search or filter functionality.
- Tool cards.
- Brief descriptions.
- Relevant featured tools.
- Useful explanatory content.
- Links to related categories.

Example:

/categories/calculators/

Calculators

Free online calculators for mathematics,

business, percentages, finance and everyday tasks.

Popular:

- Percentage Calculator

- Age Calculator

- Discount Calculator

- Average Calculator

- Date Calculator

7.3 Individual tool pages

The page template should be reusable, but the actual tool, explanations, examples and FAQs should be specific to that tool.

Suggested layout:

Breadcrumbs

Tool title

Tool introduction

Interactive tool

Results and explanations

How to use this tool

Calculation methodology

Worked examples

Practical applications

Frequently asked questions

Related tools

Footer

7.4 Mobile-first standards

A large proportion of visitors may access the website through mobile devices.

Therefore:

- Use full-width input fields on small screens.
- Avoid horizontal scrolling.
- Keep primary actions visible.
- Use readable font sizes.
- Avoid intrusive popups.
- Make results easy to copy.

- Avoid advertisements interfering with the task.

Part 8. AdSense implementation architecture

Do not integrate advertisements before the tool functionality and page layouts are stable.

The advertising system should be modular.

8.1 Advertising component

Create one reusable component for advertising locations.

```
type AdSlotProps = {
 slot: string;
 format?: "auto" | "rectangle";
};


export function AdSlot({
 slot,
 format = "auto",
}: AdSlotProps) {
 return (
  <div
   className="ad-container"
   aria-label="Advertisement"
  >
   {/* Approved AdSense unit goes here */}
  </div>
 );
}
```

This is only a layout placeholder, not a complete AdSense integration.

Once approved, use the actual ad code supplied through your AdSense account, following Google's current implementation instructions.

8.2 Advertising locations

Recommended locations to evaluate:

Location	Recommendation
Header	Avoid placing ads beside navigation
Between tool and explanations	Possible if clearly separated
Within long-form educational content	Possible if non-disruptive
Between content sections	Possible if clearly distinguishable
Beside calculator controls	Avoid
Beside download buttons	Avoid
Result area	Avoid
Footer	Optional, subject to content and layout

The exact placement must be assessed on desktop and mobile. These are design suggestions, not guaranteed approved placements.

8.3 ads.txt

When AdSense instructs you to implement an ads.txt file, publish the exact authorized seller information provided by your account.

It should be publicly accessible at:

https://yourdomain.com/ads.txt

Never invent publisher IDs or seller records.

8.4 Consent management

Build the website so that the consent solution can be integrated without rewriting every page.

For international traffic, choose a suitable Google-certified CMP and implement its documented integration.

Google's certified CMP requirements apply to relevant personalized advertising in the EEA, UK and Switzerland.

Google AdSense Help+1

8.5 Ad placement testing

Before activating ads, check:

- Desktop placement.
- Mobile placement.
- Ads beside controls.
- Ads beside download actions.
- Accidental-click risks.
- Content overlap.
- Navigation obstruction.
- Page loading behavior.
- Consent behavior.

Never make ad clicks a condition for using a tool.

Part 9. Technical SEO and traffic acquisition

AdSense monetization requires a compliant website, but sustainable revenue also requires real visitors.

We should build the traffic strategy around useful tools and search intent.

9.1 Search Engine Optimization

Set up:

- Google Search Console.
- XML sitemap.
- Robots.txt.
- Canonical URLs.
- Appropriate indexing directives.
- Breadcrumb navigation.
- Structured data where appropriate.
- Internal linking.

- Image optimization.
- Mobile usability.
- Core Web Vitals monitoring.

Google Search Central is the primary documentation source for these technical and content practices.

9.2 Keyword strategy

Each tool should target a distinct user problem.

For example:

Tool	**Search intent**
Word Counter	Count words online
Percentage Calculator	Calculate percentages
JSON Formatter	Format JSON online
Image Compressor	Compress images online
Age Calculator	Calculate age from date of birth
PDF Merger	Merge PDF documents online

Avoid creating multiple nearly identical pages to target small variations of the same search phrase.

9.3 Build supporting educational content

For every important category, create useful guides.

Examples:

Text tools

- How to count words accurately.
- Difference between words and characters.
- How reading time is calculated.

Calculators

- How percentage calculations work.
- How to calculate profit margin.
- Difference between simple and compound interest.

Developer tools

- How to format JSON.
- How to decode a JWT.
- What is Base64 encoding?

Image tools

- How to reduce image size.
- Difference between JPG and PNG.
- What is WebP?

These guides should demonstrate actual knowledge and complement the tools rather than simply repeat their titles.

9.4 Analytics

Track:

- Tool page views.
- Tool usage.
- Successful operations.
- Failed operations.
- Search usage.
- File-processing errors.
- Device categories.
- Returning visitors.
- Search traffic.
- Conversion from category pages to tool pages.

Do not record sensitive user input, uploaded documents or passwords in analytics events.

Part 10. Security, privacy and legal compliance

This is a separate workstream, not something to leave until launch.

10.1 Security checklist

Area	Required implementation
HTTPS	Valid SSL certificate
Input validation	Validate all user inputs
File uploads	Check actual file types
File processing	Apply file size and processing limits
API security	Restrict access and rate-limit requests
XSS protection	Escape and sanitize untrusted content
CSRF protection	Protect state-changing operations
Database	Restrict access and use secure credentials
Secrets	Store API keys in environment variables
Logging	Avoid logging sensitive user data
Dependencies	Regularly update vulnerable packages
Backups	Secure and test backups
Error handling	Do not expose internal server details

10.2 Privacy engineering

Our privacy architecture should follow these principles:

1. Process information locally whenever practical.
2. Collect only necessary information.
3. Explain why information is collected.
4. Avoid retaining uploaded files unnecessarily.

5. Give users appropriate privacy choices.
6. Disclose analytics and advertising technologies.
7. Provide a process for privacy-related requests.
8. Ensure third-party services are disclosed.
9. Review relevant data protection laws in the markets served.

Do not promise zero data collection or complete anonymity if your hosting provider, analytics service or advertising infrastructure processes personal data.

Part 11. A practical 90-day implementation roadmap

I would build the project in stages.

90-day development roadmap

1. **Days 1–7: Strategy and foundation**
    - Choose brand and domain.
    - Research competing websites.
    - Select the initial 30–40 tools.
    - Create site architecture.
    - Design the visual identity.
    - Set up development environment.
    - Configure hosting and deployment.
2. **Days 8–21: Core platform**
    - Build homepage.
    - Build navigation.
    - Build search.
    - Build category pages.
    - Create reusable tool components.
    - Implement first 15 tools.
    - Develop mobile-responsive layouts.
3. **Days 22–35: Complete first tool collection**
    - Implement remaining initial tools.

- Add validation and error handling.
- Test formulas and outputs.
- Add related tools.
- Develop original supporting content.
- Create tool documentation.

4. **Days 36–45: Content and trust**
    - Complete About page.
    - Complete Privacy Policy.
    - Complete Terms.
    - Complete Contact page.
    - Complete Disclaimers.
    - Establish editorial and corrections processes.
    - Review copyright and privacy practices.
5. **Days 46–55: SEO and technical quality**
    - Configure Search Console.
    - Generate sitemap.
    - Configure robots.txt.
    - Implement canonical URLs.
    - Test mobile usability.
    - Test performance.
    - Review accessibility.
    - Test security.
6. **Days 56–65: Launch and validation**
    - Launch the website.
    - Test all tools in production.
    - Check indexing.
    - Monitor errors.
    - Improve navigation.

- Publish supporting guides.

7. **Days 66–75: Monetization preparation**
    - Review all AdSense policies.
    - Implement appropriate consent management.
    - Review ad placement designs.
    - Verify site ownership.
    - Prepare AdSense application.
    - Confirm all pages and tools are complete.
8. **Days 76–90: Application and improvements**
    - Submit the site for AdSense review if ready.
    - Monitor the application.
    - Resolve any issues Google identifies.
    - Continue publishing genuinely useful tools.
    - Monitor search performance.
    - Plan the next tool collection.

These are project-planning estimates, not Google's approval timeline or an indexing guarantee.

Part 12. The complete pre-application checklist

Use this as your internal release gate.

AdSense readiness audit

0/43 completed

1. Website identity and ownership

I own or control the submitted domain.

My website has a clear identity and purpose.

My website is accessible without unnecessary login restrictions.

My contact information is functional.

My website is not impersonating another brand.

2. Content and tools

Every published tool works correctly.

Every tool provides a real and useful result.

Each tool has relevant original explanations.

Tool pages have appropriate examples.

Content is factually accurate.

There are no unfinished or placeholder pages.

There is no scraped or copied content.

There are no misleading generators or fake functionality.

3. Trust and privacy

About page is complete.

Contact page is functional.

Privacy Policy reflects actual practices.

Terms of Service are published.

Relevant disclaimers are published.

Advertising and analytics are disclosed.

Consent management is correctly configured where required.

4. Technical SEO

HTTPS works.

Sitemap is available.

Robots.txt is configured correctly.

Important pages are indexable.

Canonical URLs are correct.

Navigation and internal links work.

Mobile experience is tested.

Pages load reliably.

Google can access the content.

5. Advertising compliance

Ad placement does not encourage accidental clicks.

Ads are distinguishable from website controls.

There are no misleading ad labels.

Ads do not obstruct tool functionality.

There is no incentivized clicking.

There is no automated or fraudulent traffic.

No ads are placed on empty or low-value screens.

6. Final quality assurance

Calculations have been independently verified.

File processing has been tested.

Input validation is working.

Security checks are complete.

Analytics does not collect sensitive input.

Errors are handled properly.

The website is ready for visitors.

0% of internal checks completed

Passing this checklist does not guarantee approval. It is an internal quality-control process.

Part 13. The master prompt for building the entire platform

This is the prompt I recommend using as the project's central development instruction.

It combines the responsibilities of the full engineering and product team, while keeping AdSense compliance as a design constraint throughout development.

MASTER PROJECT PROMPT: BUILD AN ORIGINAL, ADSENSE-READY ONLINE TOOLS PLATFORM

Project objective

Act as a complete multidisciplinary product development agency responsible for researching, planning, designing, developing, testing, launching and maintaining a professional online tools platform.

The primary business objective is to create a collection of genuinely useful, free online tools that can attract organic search traffic and become eligible for Google AdSense monetization.

AdSense approval is not guaranteed. Build the platform to comply with the current Google AdSense Program Policies, Google Publisher Policies, applicable privacy requirements and Google Search spam policies.

1. Your professional roles

Work as all of the following:

1. Senior product strategist.
2. Business analyst.
3. Technical project manager.
4. Full-stack software architect.
5. Senior frontend developer.
6. Senior backend developer.
7. TypeScript engineer.
8. UI/UX designer.
9. Responsive web designer.
10. Accessibility specialist.
11. Database engineer.
12. DevOps engineer.
13. Application security engineer.
14. Technical SEO specialist.
15. Content strategist.
16. Professional technical writer.
17. Google AdSense compliance specialist.
18. Privacy engineering specialist.
19. Quality assurance engineer.
20. Performance optimization engineer.
21. Analytics specialist.
22. Conversion optimization specialist.

All roles must collaborate rather than work independently.

2. Product vision

Build one centralized website offering multiple categories of useful web-based tools.

The platform must have:

- A professional homepage.
- A searchable tool directory.
- Organized tool categories.
- Individual tool pages.
- Related tools.
- Helpful guides and educational content.
- About page.
- Contact page.
- Privacy Policy.
- Terms of Service.
- Disclaimer.
- Cookie and consent management where required.
- Editorial policy.
- Corrections process.
- Accessibility information.
- A scalable advertising integration architecture.

3. Tool categories

Develop tools in the following categories:

1. Text and writing.
2. Mathematics and calculators.
3. Finance and business.
4. Image processing.
5. PDF and documents.
6. Developer and programming.
7. Internet and networking.
8. SEO and website management.

9. Education and student utilities.
10. Everyday and lifestyle.
11. Business and productivity.
12. Security and privacy.

Begin with 30–40 highly useful tools selected for low development cost, high practical value, and manageable maintenance requirements.

Create a prioritized roadmap for the remaining tools.

Do not publish unfinished or fake tools.

4. AdSense compliance requirements

Treat AdSense compliance as a non-negotiable engineering requirement.

Before development, examine the latest official documentation covering:

- AdSense eligibility.
- AdSense Program Policies.
- Google Publisher Policies.
- Low-value content restrictions.
- Site behavior.
- Invalid traffic.
- Ad placement.
- Privacy disclosures.
- Consent requirements.
- Google Search spam policies.
- Scaled content abuse.
- Helpful, people-first content.

Use official Google documentation as the primary authority.

Never claim that a particular tool category or number of pages guarantees approval.

Do not:

- Publish empty or unfinished pages.
- Generate large numbers of repetitive pages.

- Copy competitors' content.
- Use scraped articles.
- Build misleading generators.
- Encourage ad clicks.
- Put advertisements beside action buttons in a way that risks accidental clicks.
- Place ads on empty or error screens.
- Use bots or fraudulent traffic.
- Misrepresent functionality.
- Conceal advertising as website navigation.

Every published tool must provide genuine functionality and meaningful visitor value.

5. Technology requirements

Recommended architecture:

- Next.js.
- TypeScript.
- Tailwind CSS.
- Reusable React components.
- Accessible UI components.
- PostgreSQL when persistent storage is necessary.
- Supabase when managed database services are useful.
- Vercel or equivalent hosting.
- Browser-side processing wherever practical.

Use appropriate specialized libraries for PDF, image, QR code, document and programming utilities.

Do not introduce unnecessary infrastructure.

6. UI/UX requirements

Design a professional, responsive and consistent design system.

Every tool must provide:

- Clear title and description.

- Accessible input fields.
- Instructions.
- Primary action.
- Output/result area.
- Error handling.
- Reset functionality where applicable.
- Copy functionality where applicable.
- Download functionality where applicable.
- Mobile compatibility.
- Keyboard accessibility.
- Relevant examples.
- Related tools.

Advertisements must never interfere with primary functionality.

7. SEO and content requirements

Every tool page must have:

- Unique page title.
- Unique meta description.
- Descriptive URL.
- Appropriate heading hierarchy.
- Original explanatory content.
- Instructions.
- Calculation methods or processing explanations.
- Real examples.
- Limitations.
- Relevant FAQs.
- Internal links.
- Appropriate canonical URLs.
- Indexability controls.

Never use arbitrary word counts as a substitute for content quality.

Do not automatically publish AI-generated pages without editorial review, accuracy checks and meaningful original value.

8. Security and privacy requirements

Implement:

- HTTPS.
- Secure input validation.
- XSS protection.
- CSRF protection where applicable.
- Secure API handling.
- Rate limiting.
- Secure environment variables.
- File validation.
- Temporary file handling.
- Automatic deletion of temporary uploads.
- Appropriate logging controls.
- Data minimization.
- Transparent privacy disclosures.
- Suitable consent management.

Never claim that user data is private, anonymous or deleted unless the technical implementation supports that claim.

9. Tool development workflow

For every tool:

Step 1: Research the problem and user requirements.

Step 2: Write a product requirements document.

Step 3: Define the functionality and edge cases.

Step 4: Design desktop and mobile interfaces.

Step 5: Select the appropriate processing technology.

Step 6: Implement the tool.

Step 7: Validate calculations and results.

Step 8: Write original explanatory content.

Step 9: Implement technical SEO.

Step 10: Perform security and accessibility testing.

Step 11: Review AdSense suitability.

Step 12: Publish only after passing quality checks.

10. Quality standards

The website must:

- Work on mobile, tablet and desktop.
- Have consistent design.
- Load efficiently.
- Provide correct results.
- Handle invalid inputs.
- Be accessible.
- Be secure.
- Be easy to navigate.
- Avoid intrusive advertising.
- Provide genuine user value.

Test every tool against normal inputs, edge cases, invalid inputs and unexpected conditions.

11. Project execution

Do not attempt to build every tool simultaneously.

Divide the work into milestones:

Milestone 1: Research and architecture.

Milestone 2: Design system and homepage.

Milestone 3: Reusable tool engine.

Milestone 4: First 30–40 tools.

Milestone 5: Original content and trust pages.

Milestone 6: SEO and accessibility.

Milestone 7: Security and quality assurance.

Milestone 8: AdSense readiness review.

Milestone 9: Launch and monitoring.

Milestone 10: Expansion into additional categories.

For each milestone, provide:

- Tasks.
- Deliverables.
- Files to create.
- Dependencies.
- Acceptance criteria.
- Testing instructions.
- Risks.
- Estimated development effort.

12. Working instructions

First produce a comprehensive project requirements document and prioritized development roadmap.

Then establish the application architecture and design system.

Build the application in functional stages.

Provide complete working source code, not pseudocode or empty placeholders for core functionality.

When a tool is completed, demonstrate how it works and document how it was tested.

Keep a list of completed, pending and blocked features.

Do not invent credentials, API keys, advertising identifiers or legal claims.

Do not submit the website to AdSense without the owner's explicit authorization.

Your final deliverable must be a production-ready, maintainable online tools platform with a documented AdSense compliance and quality assurance process.

Part 14. My final strategic recommendation

There is one decision that will significantly affect the success of this project: whether to build a general-purpose tools website or a more specialized platform.

I recommend a broad tools platform with a strong initial focus on text, calculators, developer utilities and everyday conversions.

Recommended launch portfolio

12

Text tools

Writing and text processing

10

Calculators

Mathematics and everyday calculations

10

Developer tools

Programming and data utilities

8

Everyday tools

Conversions and productivity

Initial target

40 tools

A proposed development scope, not an AdSense eligibility threshold.

Part 15. Let's establish the actual project before coding

I want to make sure we build the correct business rather than just produce a collection of disconnected tools.

There is one important decision to make first.

Project setup

1. What type of platform do you want?

All-in-one platform with every category

Developer, SEO and technical tools

Calculators, finance and everyday tools

Start with the easiest and most profitable opportunities

2. What is your development budget?

As close to zero cost as possible

Low budget, with essential paid services allowed

Moderate budget for professional infrastructure

Flexible budget for the best solution

3. What is your coding experience?

Beginner – I need you to guide everything

I know a little coding

I can develop and deploy applications

I prefer no-code or AI-assisted development

4. Where should we host it?

Choose the best affordable hosting for me

Vercel

WordPress

Another platform

5. How much should we build in the first development stage?

Complete platform, developed in manageable stages

First 40 tools, then expand

10 excellent tools, validate the business, then expand

What happens next: I will turn your choices into the actual project specification, domain and brand strategy, complete folder structure, homepage design, reusable tool architecture, first tool implementations, original content plan, and AdSense compliance testing process.

The aim is not simply to get a website online. It is to create a sustainable web-tools business that is genuinely useful, technically reliable, search-friendly and eligible to apply for AdSense.

Official Google documentation to keep as our source of truth

1. AdSense eligibility requirements

2. Prepare your website for AdSense
3. AdSense Program Policies
4. Google Publisher Policies
5. Invalid traffic guidance
6. Ad placement policies
7. Helpful, reliable, people-first content
8. Google Search spam policies
9. Google-certified consent management requirements
10. AdSense site connection and approval process

These documents should be revisited before launch because Google's requirements and enforcement guidance can change.

Your project is now defined: an all-in-one free online tools platform

Project configuration established

Zero-budget architecture

We will build the complete platform, not just a demonstration or a collection of mockup pages.

Your decisions establish the following requirements:

Decision	Project specification
Platform	All-in-one tools platform
Budget	$0 initial infrastructure budget
Coding experience	Beginner-friendly
Hosting	Best suitable free hosting
Development scope	Complete platform, delivered in stages
Monetization	Google AdSense, subject to approval
Primary market	International
Development approach	AI-assisted, modular, scalable

1. The product we're building

I built 200+ free online tools — here's everything I learned in 8 months - DEV Community

dev.to

Toolora

Your proposed brand name — Free Online Tools for Everyday Work

One platform for calculations, text processing, programming, images, documents, SEO, business, education and everyday productivity.

12 categories

271 planned tools

Mobile-first

AdSense-conscious

Toolora is a working project name, not a confirmed available domain or trademark. We will check availability before committing to it.

The core business model:

1. Visitors find individual tools through Google.
2. They use the tools without paying or registering.
3. They discover related tools and useful guides.
4. We build genuine repeat traffic and search visibility.
5. Once the website meets the requirements, we apply for AdSense.
6. If approved, we introduce compliant advertising.

Important: We will not build the website around advertisements. We will build it around useful tools, with advertising as a monetization layer.

Part 16. Important infrastructure decision: truly zero-budget hosting

I have checked the current hosting terms, and there is an important issue with the original hosting recommendation.

I would not choose Vercel Hobby for this monetized project. Its published terms restrict the Hobby plan to personal, non-commercial use. An AdSense business is a commercial project.

Vercel+1

Our hosting decision must consider the actual commercial terms, not just whether the hosting company advertises a free tier.

My recommended architecture is:

Service	Purpose	Initial approach
Cloudflare Pages	Website hosting	Free tier, subject to commercial-use terms
GitHub	Source code and version control	Free repository
Next.js	Application framework	Open source
TypeScript	Programming language	Free
Tailwind CSS	Styling	Free
Browser APIs	Calculations and image processing	Free
Google Search Console	Search monitoring	Free
Google Analytics	Traffic measurement	Free option
AdSense	Advertising	No initial subscription
Domain	Custom website address	Requires checking cost

Cloudflare Pages is my proposed hosting candidate, but we must verify the current applicable usage limits and terms before deployment. Free hosting does not mean unlimited processing or guaranteed availability.

One unavoidable business cost: a custom domain normally requires payment. We can begin on a free provider subdomain, but I recommend purchasing a domain when the budget allows. A free subdomain is not automatically disqualified from AdSense; eligibility depends on the site and the relevant hosting arrangement.

Google AdSense Help

For now, we will build the application so that it can be moved between compatible hosts.

Part 17. Stage 1 implementation: the actual application

We will use Next.js 16 with TypeScript, with a static-export-compatible architecture for the initial release. Next.js 16 is currently the active major version, and its installation documentation supports the modern App Router, TypeScript and Tailwind setup.

Next.js+2

One additional hosting qualification: Cloudflare's published free plans are described as intended for personal or hobby projects, and its Pages documentation specifies free-tier build and file limits. We should verify the applicable terms for your commercial use before committing to it.

cloudflare.com+1

What I am implementing first

Stage 1 — Working foundation

Implementation scope

- Professional homepage.
- Responsive navigation.
- Searchable tool directory.
- Twelve tool categories.
- Individual tool interfaces.
- Eight initial working utilities.
- About, contact, privacy, terms and disclaimer pages.
- SEO metadata and sitemap.
- Advertising placeholders, disabled by default.
- A beginner-friendly installation guide.

The initial eight tools will be:

1. Word Counter.
2. Case Converter.
3. Duplicate Line Remover.
4. Text Reverser.
5. Percentage Calculator.
6. Age Calculator.
7. JSON Formatter and Validator.
8. Base64 Encoder and Decoder.

The first release will process these tools locally in the browser wherever possible. This means no accounts, no paid APIs, no database subscription and no unnecessary storage of user input.

Download the implementation

Toolora — Stage 1 source code

ZIP archive · Next.js · TypeScript · CSS

Created

Download the complete starter project

Project Requirements Document

Scope, architecture, compliance, acceptance criteria and roadmap.

Download the project requirements

Installation and setup guide

Read the README

Part 18. How to start the application

You don't need to know programming to get the initial application running.

Installation instructions

1. Install Node.js

Go to Node.js official website and install the current supported version. The project requires Node.js 20.9 or newer.

2. Install Visual Studio Code

Download Visual Studio Code. This is where you will edit your website.

3. Extract the ZIP file

Extract toolora-starter.zip into a folder on your computer.

4. Open the project

Open the extracted toolora-starter folder in Visual Studio Code.

5. Open the terminal

Select Terminal → New Terminal.

6. Install dependencies

Run:

npm install

7. Start the website

Run:

npm run dev

8. Open your website

Visit:

http://localhost:3000

Build verification pending

I created the project files and verified the ZIP archive. However, the package installation attempt timed out in this environment, so I have not yet verified a successful Next.js production build. Run the installation and build locally before deploying.

Part 19. The remaining development stages

We have established the foundation. Here is how I intend to complete the larger platform.

Stage	Deliverable	Status
1	Homepage, architecture, eight working tools, initial pages	Files created
2	Expand to 40 functional tools	Next
3	Individual content, examples and tool documentation	Pending
4	Image and PDF processing	Pending
5	SEO, networking and advanced utilities	Pending
6	Privacy, accessibility, security and performance audit	Pending
7	Hosting, domain, Search Console and launch	Pending
8	AdSense readiness review and application	Pending

My non-negotiable rules for this project

1. No fake tools.
2. No copied or spun content.
3. No advertisements before approval.
4. No unnecessary paid APIs.

5. No unnecessary collection of visitors' information.
6. No unfinished pages presented as working products.
7. No mass-produced thin SEO pages.
8. Every tool must be tested before publication.

Google specifically identifies misleading functionality and scaled content abuse as Search spam violations.

Google for Developers

The initial privacy and legal pages in the ZIP are development templates, not finished legal documents. We will complete them based on your actual data practices.

Three things we must resolve before public launch

- Brand: Confirm Toolora or choose another name.
- Domain: Choose a domain and configure the canonical URL. A custom domain will normally require some payment.
- Hosting: Confirm a provider's commercial-use terms. Vercel Hobby is restricted to non-commercial use, while Cloudflare's free plan is described as intended for personal or hobby projects. We should not assume either is suitable for an AdSense business without checking its applicable terms.

Vercel+1

What I want to accomplish in Stage 2

My next development target is to bring the application to 40 functional tools, not simply display 40 cards.

That means implementing the actual calculation and processing engines, validating results, writing the individual tool explanations, and testing all the functionality.

The guiding principle: 40 genuinely useful tools are better than 271 unfinished pages.

Once the first 40 are stable, we can expand the other categories without compromising quality.

Your project source code and requirements document are ready for you to begin locally. We can continue building the next stage directly from this foundation.

PROJECT_REQUIREMENTS.mdDocumentREADME.mdDocument

Toolora — online tools starter

A beginner-friendly Next.js starter for an all-in-one online tools website. The initial release includes eight browser-based utilities and a directory of planned tools. Planned tools are clearly marked and do not pretend to work.

Important before launch

- Replace every `example.com` value with the real canonical domain in `app/layout.tsx`, `app/robots.ts`, and `app/sitemap.ts`.
- Confirm the brand name and check trademark/domain availability.
- Replace the contact page's instructions with a real monitored contact address.
- Rewrite and legally review the privacy policy, terms and disclaimer for the actual operator, jurisdiction, hosting, analytics, advertising, cookies, consent and data retention practices.
- Do not enable AdSense until the site is approved and the supplied ad code, publisher ID and slot IDs are correctly configured.
- Do not treat the included policy templates as legal advice or complete compliance.
- Verify hosting terms permit your intended commercial/advertising use. Vercel Hobby is not appropriate for a commercial monetized project under its published terms. Check Cloudflare's current terms and plan fit before deploying.

Requirements

- Node.js 20.9 or newer
- npm

Run locally

1. Install Node.js from https://nodejs.org/.
2. Extract this project folder.
3. Open a terminal in the `toolora-starter` folder.
4. Run `npm install`.
5. Run `npm run dev`.
6. Open http://localhost:3000.

Production build

Run `npm run build`. This project uses Next.js static export (`output: 'export'`) and writes static output to `out/`.

Deploying a static export

Connect the Git repository to a static host that supports Next.js static exports. Build command: `npm run build`. Output directory: `out`. Check the host's current terms and limits before using it for a monetized website. Dynamic APIs, server-side rendering, server uploads and database-backed features are not included in this static first stage.

Current live tools

- Word Counter
- Case Converter
- Duplicate Line Remover
- Text Reverser
- Percentage Calculator
- Age Calculator
- JSON Formatter & Validator
- Base64 Encoder & Decoder

Roadmap

The directory includes planned examples across 12 categories. Planned tools must be implemented, documented, tested and reviewed before their status is changed to `live`.

AdSense safety

There are no active ads or ad code in this starter. Keep ads disabled until approval. When approved, integrate Google's actual code as directed by the account and policy documentation. Keep ads clearly distinct from controls and never encourage clicks. Implement an appropriate Google-certified CMP for relevant EEA/UK/Swiss personalized ad traffic. Review official policies at https://support.google.com/adsense/answer/48182 and https://support.google.com/adsense/answer/13554116.

Known limitations in this first stage

- No user accounts or database.
- No server-side file uploads or PDF processing.
- No external live data APIs.

- Tool content is a starting editorial draft and must be fact-checked and improved before public launch.
- The starter privacy and legal pages are incomplete templates, not publish-ready documents.
- The age calculator uses the device's local date/time interpretation; test date edge cases and timezone behavior before launch.
- Word and sentence counting rules are intentionally simple and should be documented as approximate.